The L_{Var} language (Project 2)

CIS531 — Fall 2025, Syracuse

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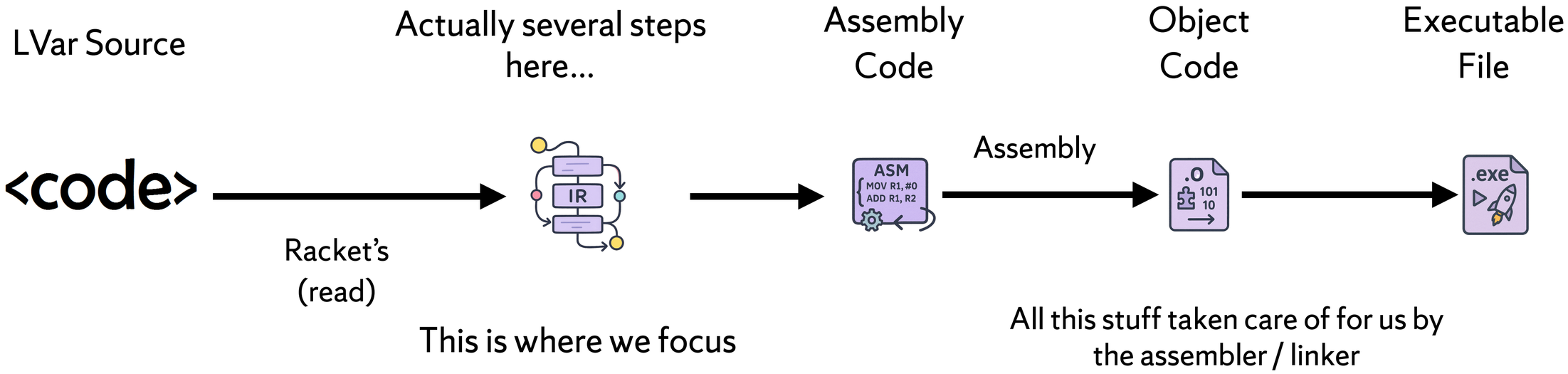
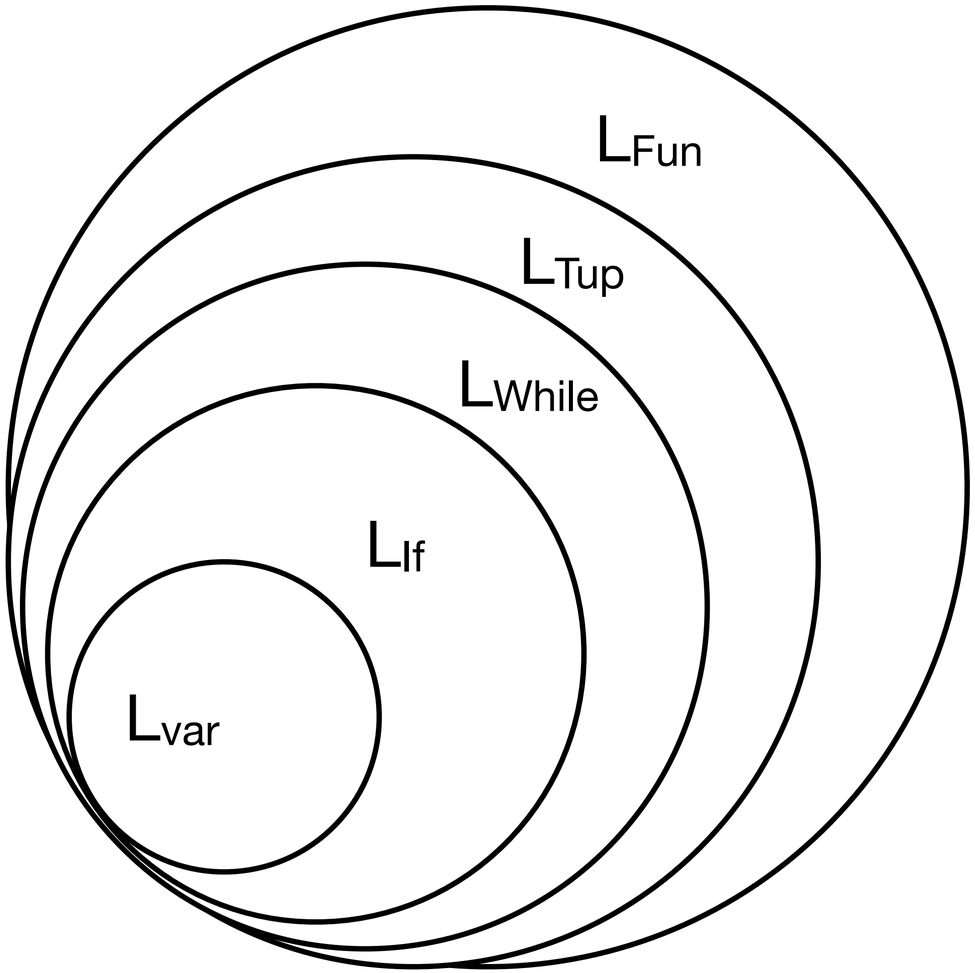
Project 2 in CIS352

- This week we'll dive into Project 2 in CIS352
- You'll write a compiler for LVar, a language that extends R0 to several other forms
- You'll also compile LVar to x86_64 assembly and verify it works
- Project due in roughly 14 days



The LVar language



We cover LVar, the first language in the stack

Order of language the book presents:

- **LVar — integer +/-, straight-line variables**
- LIf — branching control
- LWhile — loops
- LTup — heap-allocated data, GC
- LFun — functions
- Some more…

The LVar (R1) language (Grammar)

```
exp ::= int | (read) | (- exp) | (+ exp exp)
          | var    | (let ([var exp]) exp)

R1 ::= (program exp)
```

Notice that compared to R0, R1 adds **variables** and **let-binding**
Handling naming, shadowing, etc. will present novel challenges for us to solve

Livecoding: let's write an interpreter for LVar / R1

In-class demo…

The x86 language

This chapter covers a specific subset of x86_64:

- Only 64-bit integers
- Limited addressing modes
- Only ***eight*** instructions (plus labels)
- Whole program is just consecutive instructions in one function: main

reg	::=	`rsp` | `rbp` | `rax` | `rbx` | `rcx` | `rdx` | `rsi` | `rdi` |
		`r8` | `r9` | `r10` | `r11` | `r12` | `r13` | `r14` | `r15`
arg	::=	`$`*int* | `%`*reg* | *int*`(%`*reg*`)`
instr	::=	`addq` *arg*, *arg* | `subq` *arg*, *arg* | `negq` *arg* | `movq` *arg*, *arg* |
		`callq` *label* | `pushq` *arg* | `popq` *arg* | `retq` | *label*`:` *instr*
prog	::=	`.globl main`
		`main:` $instr^+$

Figure 2.3: A subset of the x86 assembly language (AT&T syntax).

x86: what you need to know for P1

- Assembly code consists of **instruction sequences**
  - Grouped into “functions” (procedures)

- Each instruction does a very simple task (add, multiply, jump)

- There are a limited number of variables (registers)
  - x86-64 has 16 of these! 2 hold pointers to stack (rsp/rbp)

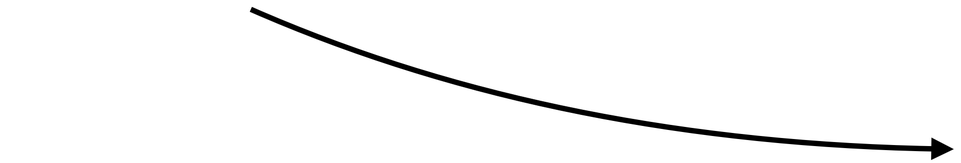
- If you need more memory (e.g., for storing an array), must store in stack / heap / etc…

- We avoid “functions” for now—mostly

x86_64 is huge, but you only need to know a little bit...

- Ugly instruction set: decades of development and extensions
  - x86_64 is the exemplar “CISC” ISA
- In practice, you will deal with a very small subset of x86_64
  - Nearly half the instruction space (by encoding) is AVX/SIMD!
- For a while, we will focus on compiling only a single function

This program should print out 500 to stdout




```
clang -arch x86_64 -o ex1 ex1.s runtime.c
```

```
    .text
    .globl main
   .extern print_int64

main:
    push    %rbp
    mov     %rsp, %rbp
    movq    $500, %rdi
    call    print_int64
    leave
    ret
```

(On Mac: make sure to prefix all symbols with _, i.e., _main and _print_int64)

```
    .text
    .globl main
   .extern print_int64

main:
    push   %rbp
    mov    %rsp, %rbp
    movq   $42, %rdi
    call   print_int64
    leave
    ret
```

Commands starting with dots (.) are directives that tell the assembler how to lay out your program

```
    .text
    .globl main
   .extern print_int64

main:
    push   %rbp
    mov    %rsp, %rbp
    movq   $42, %rdi
    call   print_int64
    leave
    ret
```

The OS loader **assumes** your binary will have a function named main (Linux ELF) or (on Mac) _main

While Linux / Mac are very close, there is one very key difference: labels in the Mach ABI use **underscores**

So printf is _printf, main is _main, etc.

```
    .text
    .globl main
   .extern print_int64

main:
    push   %rbp
    mov    %rsp, %rbp
    movq   $42, %rdi
    call   print_int64
    leave
    ret
```

Briefly, here's what the snippet does:

- **Function prelude**:
  - Each function needs to set things up
  - In this case, we push the "base pointer" %rbp
  - Then, move the "stack pointer" into %rbp
    - (Source then target in GAS Assembler)
- **Function body**:
  - We move the literal value 42 (decimal) into %rdi
    - %rdi is a general-purpose register—we'll cover this soon
    - First argument to each function is passed in %rdi
  - Next, we call print_int64 (arg in %rdi)
- **Function epilogue**:
  - Each function needs to clean things up
  - leave pops %rbp, return exits from the function

Cross-compiling on a Mac (M1/2/3/...)

- Remember, if on a Mac ***prefix all symbols with _***
- We use clang's *cross-compiler* to build a Mach-O x86_64 object file
- Then, we use clang to link the object (.o) file into a Mach-O executable
- Finally, we run it: Rosetta loads it and translates it on-the-fly to run on ARM!

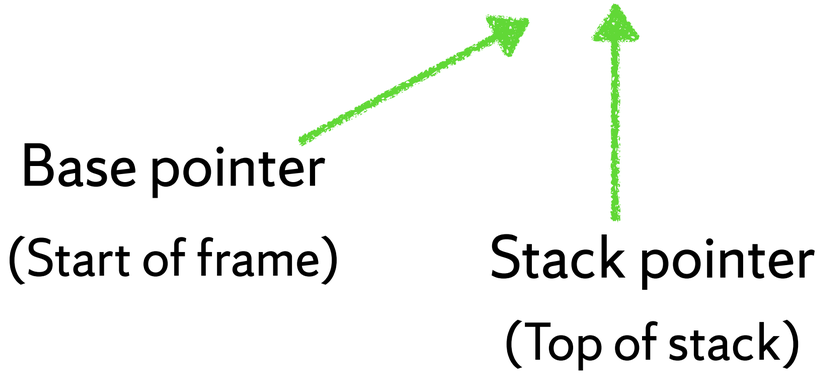
```
kkmicins@laptop ∂ → clang -target x86_64-apple-darwin -c ex0.s
kkmicins@laptop ∂ → clang -target x86_64-apple-darwin ex0.o -o prog
kkmicins@laptop ∂ → ./prog
Hello, world!
```

Registers: very fast, on-chip data

Originally, 8-bit registers: al, bl, cl, dl

Historically, x86 architectures only had **four** 16-bit general purpose registers: ax, bx, cx, dx

Also other registers: bp, sp, di, si



Registers: very fast, on-chip data

- x86-64: 16 “general purpose” registers: rax, rbx, rcx, rdx, rsp, rbp, rsi, rdi, R8-R15

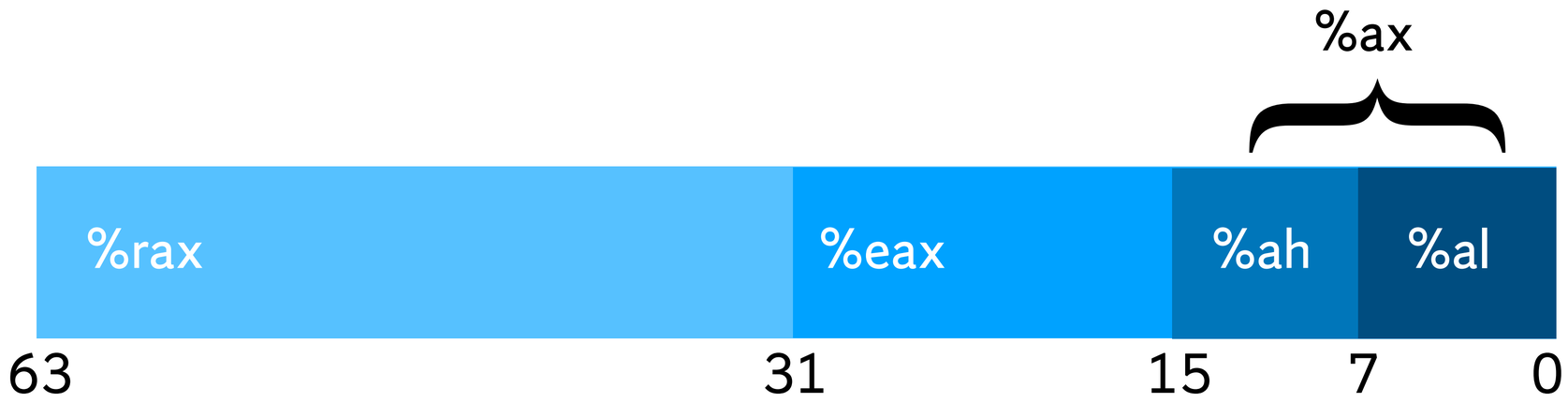
- Also some special registers you can’t modify directly: instruction pointer, flags
- Registers are a ***precious commodity***: you want to store temporaries in registers
  - Historically: ensuring something certain values kept in registers was a reason to write manually-tuned Assembly code (not as common in modern times)

Preview: register allocation (towards the end of the course…)

- You will eventually **run out** of registers, at which point you **have** to “spill” to the stack
  - Stack stores local values: medium-sized chunk of memory that you can push/pop
  - Push on function entry, pop on function exit—local values disappear once function done

Smaller registers are *nested* inside bigger ones

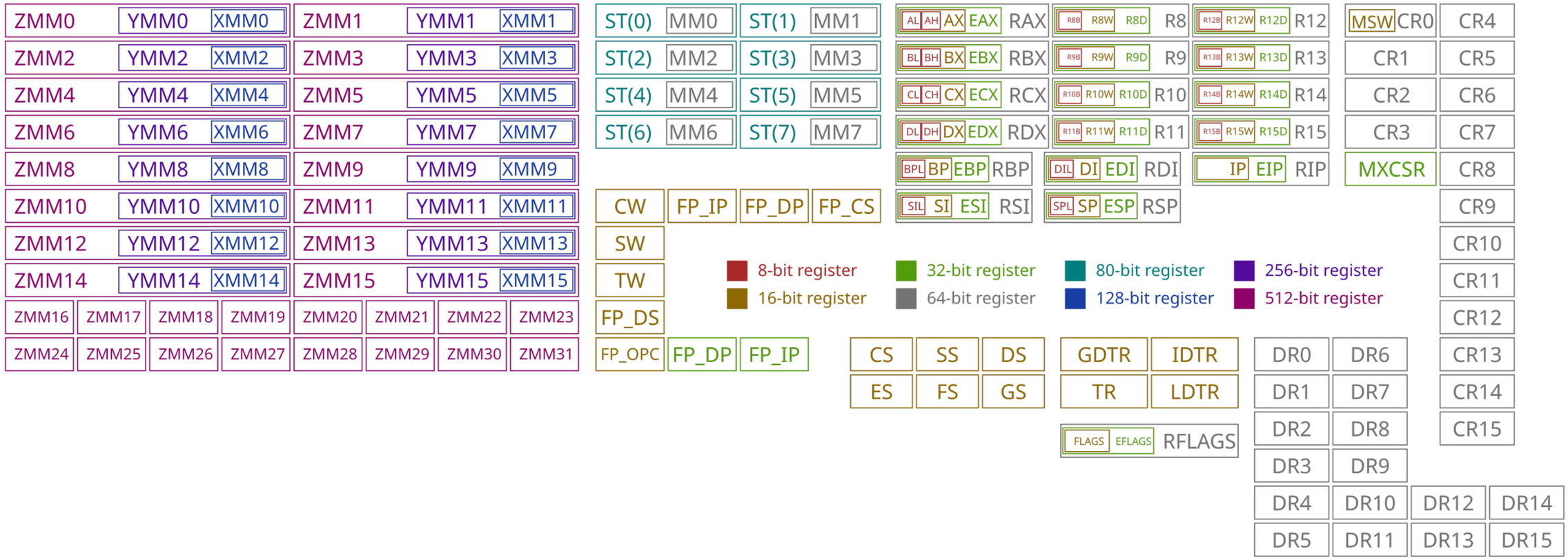
- %rax is one of the most common general-purpose registers
- But, if you use %eax, you're **really** using the ***lower 32 bits*** of %rax
- In this class, we'll make it easy: everything is 64 bits!



Registers in x86_64

- Today we mostly have 64-bit ISAs
- 32-bit registers still used (e.g., int in C++ on my machine)
- 16-bit basically gone, 8-bit totally gone
  - Some code out there that still works with the 32/16/8 bit registers, though
- Assembly instructions have **suffixes**: denote which bit-width they're working on
  - E.g., movq says "move into a **quadword** (8-bytes)"
    - "Word" is ambiguous ("machine word" is the native register size)
  - movl says "move into a **long** (4-bytes)"
  - We will try to keep things simple by working with 64-bit values values!

x86_64 register map



https://commons.wikimedia.org/wiki/File:Table_of_x86_Registers_svg.svg

Exercise:

```
    .text
    .globl main
   .extern print_int64

main:
    push   %rbp
    mov    %rsp, %rbp
    …
    # put something into %rdi
    call   print_int64
    leave
    ret
```

Translate the following into pseudocode:

x = 40

y = 20

z = 2+x * y

Using the registers %rax, %rbx, %rcx, %rdx (if needed…)

And the instructions:

add %src, %dst # %dst = %src + %dst

imul %src, %dst # %dst = %src * %dst

mov %src, %dst # put result into %rdi

Ultimately, we will sometimes run out of registers…

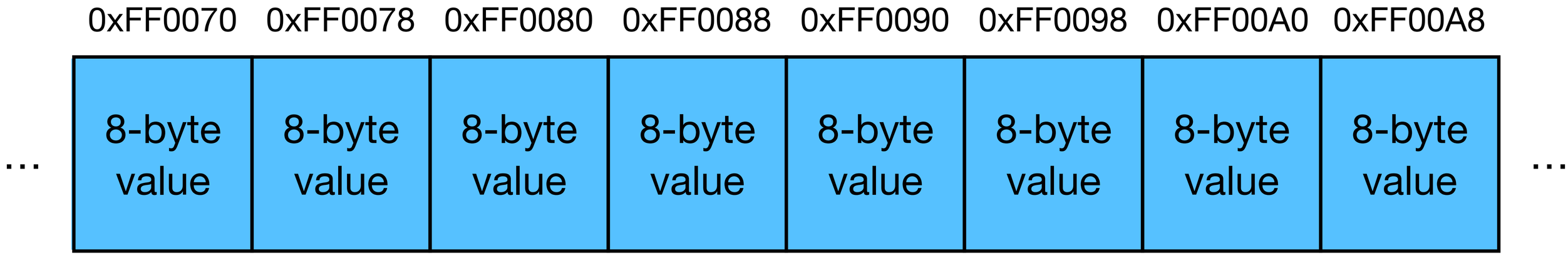
- If we only have N registers, we can only write programs that use N*8 bytes!
- Instead, also keep data in **memory**
  - When we run out of registers, use RAM

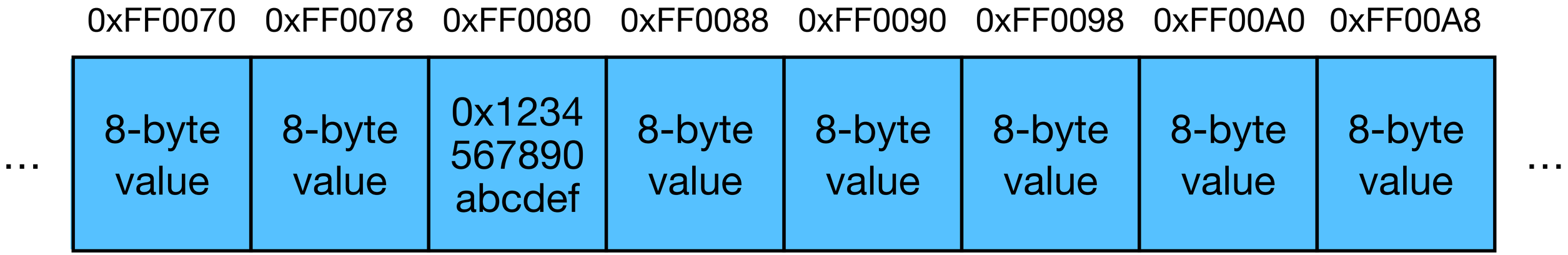
- But ultimately, most instructions ***require*** at least one operand to be a register
  - Solution: registers track “current” data, contents can be “spilled to” the stack

- Memory loads / stores ***dominate the execution time of most modern apps***
  - Most chips will have a hierarchy of caches
  - Recently-used data will generally be in the cache—not full trip to RAM

Memory: a big sea of bytes

From the POV of an assembly, we can think of memory as being a “giant array”





We can **load** from memory and **store** to memory using mov

We need a ***pointer*** to some value stored in a register...

For example, let's say we wanted to load 0xFF0080 into %rcx, assuming that %rax = 0xFF0080

```
// in C: %rcx = *%rax
movq (%rax), %rcx
```

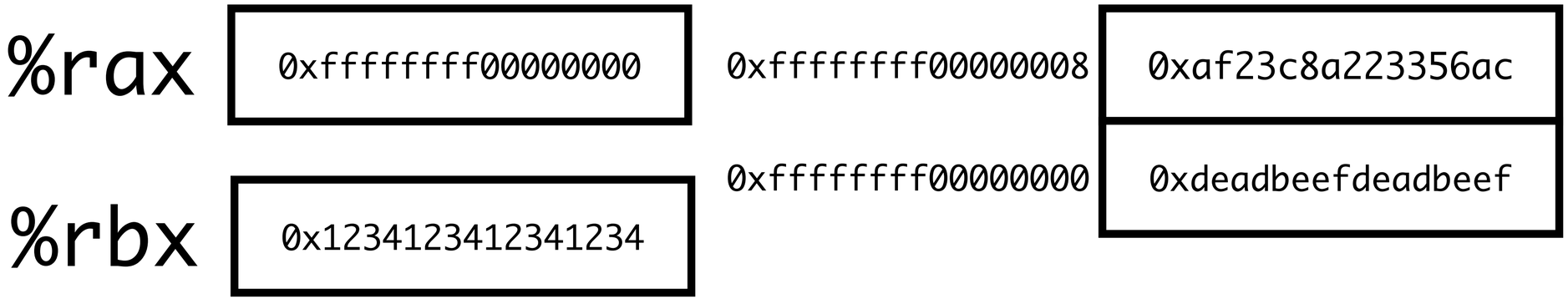
“Move the value **at address** %rax into register %rbx”

Opcode name

Destination

```
mov   (%rax), %rbx
```

Source



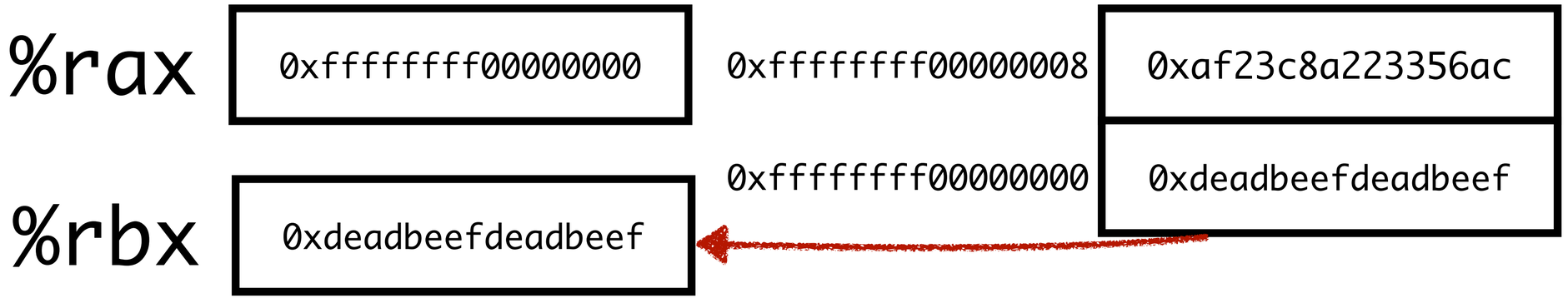
“Move the value **at address** %rax into register %rbx”

Opcode name

Destination

```
mov   (%rax), %rbx
```

Source



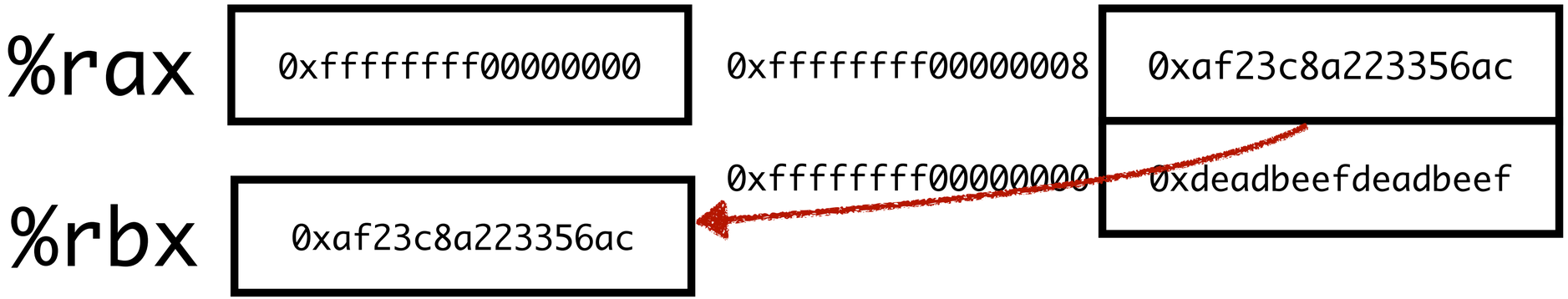
“Move the value **at address** %rax+8 into register %rbx”

Opcode name

Destination

```
mov   8(%rax), %rbx
```

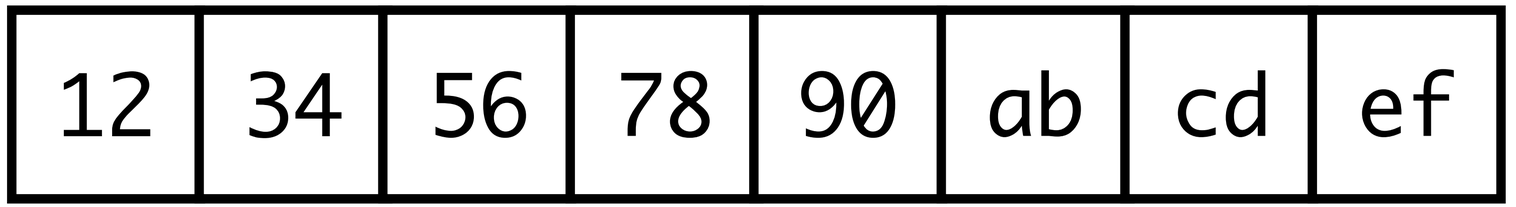
Source



Endianness

- x86 is a **little-endian** architecture
- If an n-byte value is stored at addresses a to a+(n-1) in memory, byte a will hold the least significant byte
- Note: values ***within*** bytes *do not* change—only the **byte order** changes
- How would 0x1234567890abcdef be stored in memory on a big/little-endian machine, starting at address 0xAA00?

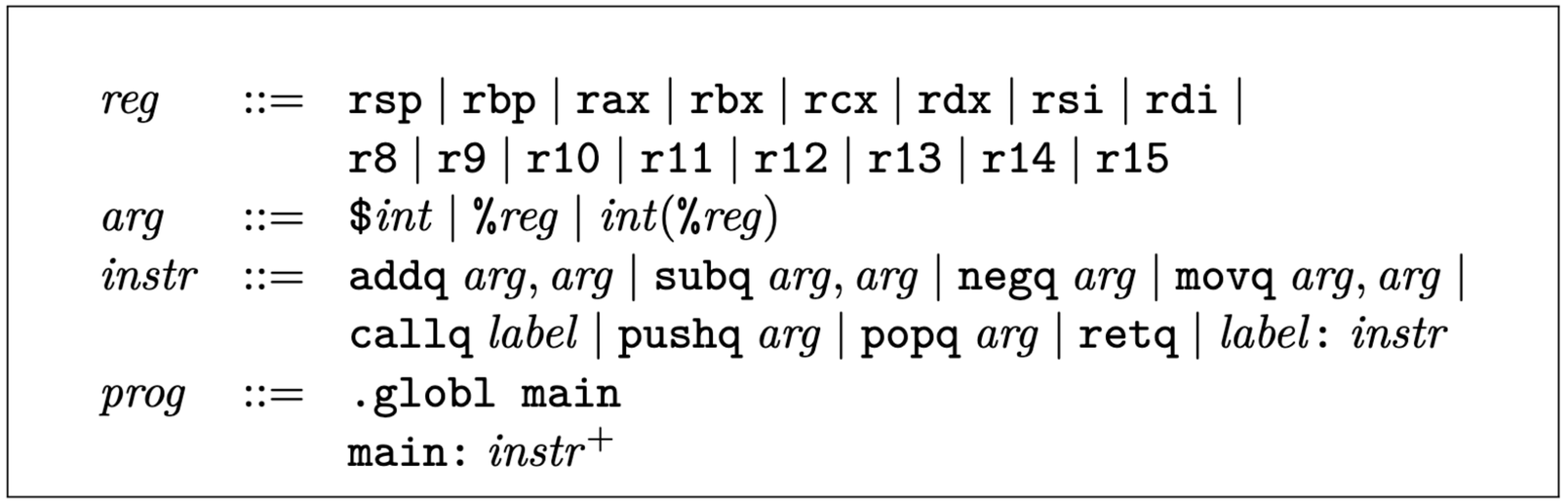
To represent `0x1234567890abcdef`



Most Significant Byte

Least Significant Byte

The only instructions you need to know for P2



reg	::=	`rsp` | `rbp` | `rax` | `rbx` | `rcx` | `rdx` | `rsi` | `rdi` |
		`r8` | `r9` | `r10` | `r11` | `r12` | `r13` | `r14` | `r15`
arg	::=	`$`*int* | `%`*reg* | *int*(`%`*reg*)
instr	::=	`addq` *arg*, *arg* | `subq` *arg*, *arg* | `negq` *arg* | `movq` *arg*, *arg* |
		`callq` *label* | `pushq` *arg* | `popq` *arg* | `retq` | *label*`:` *instr*
prog	::=	`.globl main`
		`main:` *instr*$^+$

Figure 2.3: A subset of the x86 assembly language (AT&T syntax).

- Arithmetic: **addq**, **subq**, **negq** (destination last, source first)
- Data movement: **movq**, **pushq**, **popq**
- Function call / return: **callq** , **retq**

Boilerplate code for P2

```
exp ::= int | (read) | (- exp) | (+ exp exp)
          | var    | (let ([var exp]) exp)

R1 ::= (program exp)
```

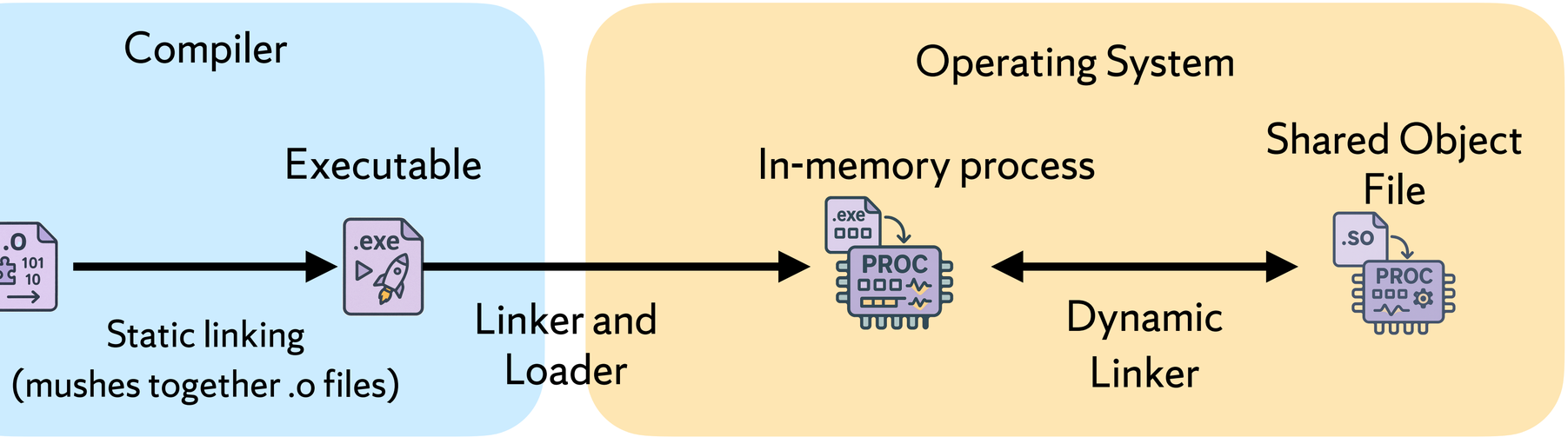
```
    .text
    .globl main
   .extern print_int64

main:
    push   %rbp
    mov    %rsp, %rbp
    <GENERATED CODE HERE>
    # Leave return value in %rdi
    call   print_int64
    leave
    ret
```

The crux of our effort will be to translate expressions into assembly language code that implements desired behavior

Leave result of *exp* in register %rdi

What happens *after* the executable is generated

- The executable is written in a very specific format which the OS understands
  - Examples: Windows Portable Executables (PE .exes), Linux ELF, Mac Mach-O, other lesser-known
- The OS typically organizes process memory into ***segments***
- The loader (roughly: what happens after execve) knows how to populate the processes memory from the .exe file. The loader and dynamic linker collaborate to load the .exe into memory
- At runtime, some calls to externally-defined functions may need to be resolved by the dynamic linker—this is accomplished by some technical systems tricks which I will not explain in detail now



ELF[101] a Linux executable walkthrough

ANGE ALBERTINI
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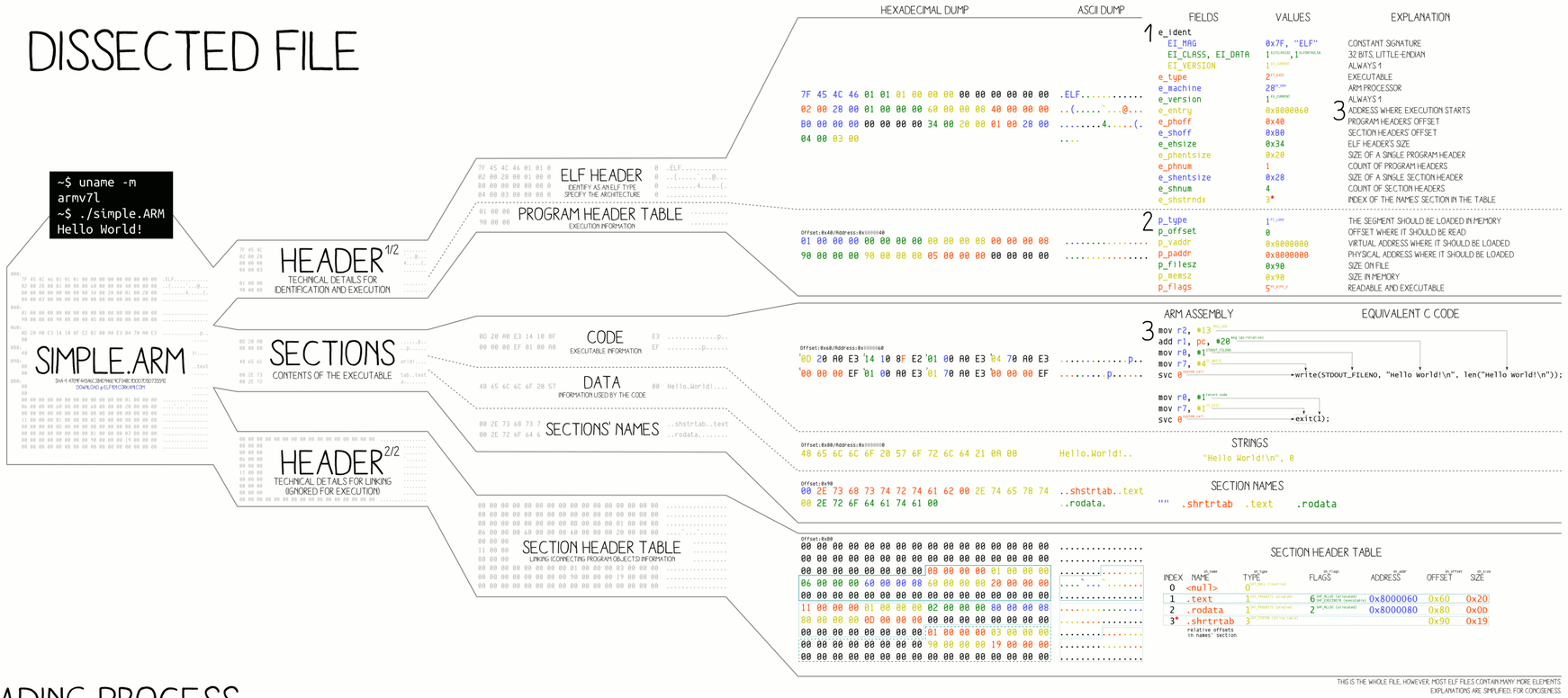
DISSECTED FILE



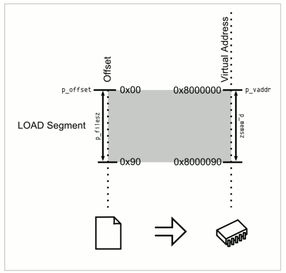
LOADING PROCESS

1 HEADER

THE ELF HEADER IS PARSED
THE PROGRAM HEADER IS PARSED
(SECTIONS ARE NOT USED)

2 MAPPING

THE FILE IS MAPPED IN MEMORY
ACCORDING TO ITS SEGMENT(S)



3 EXECUTION

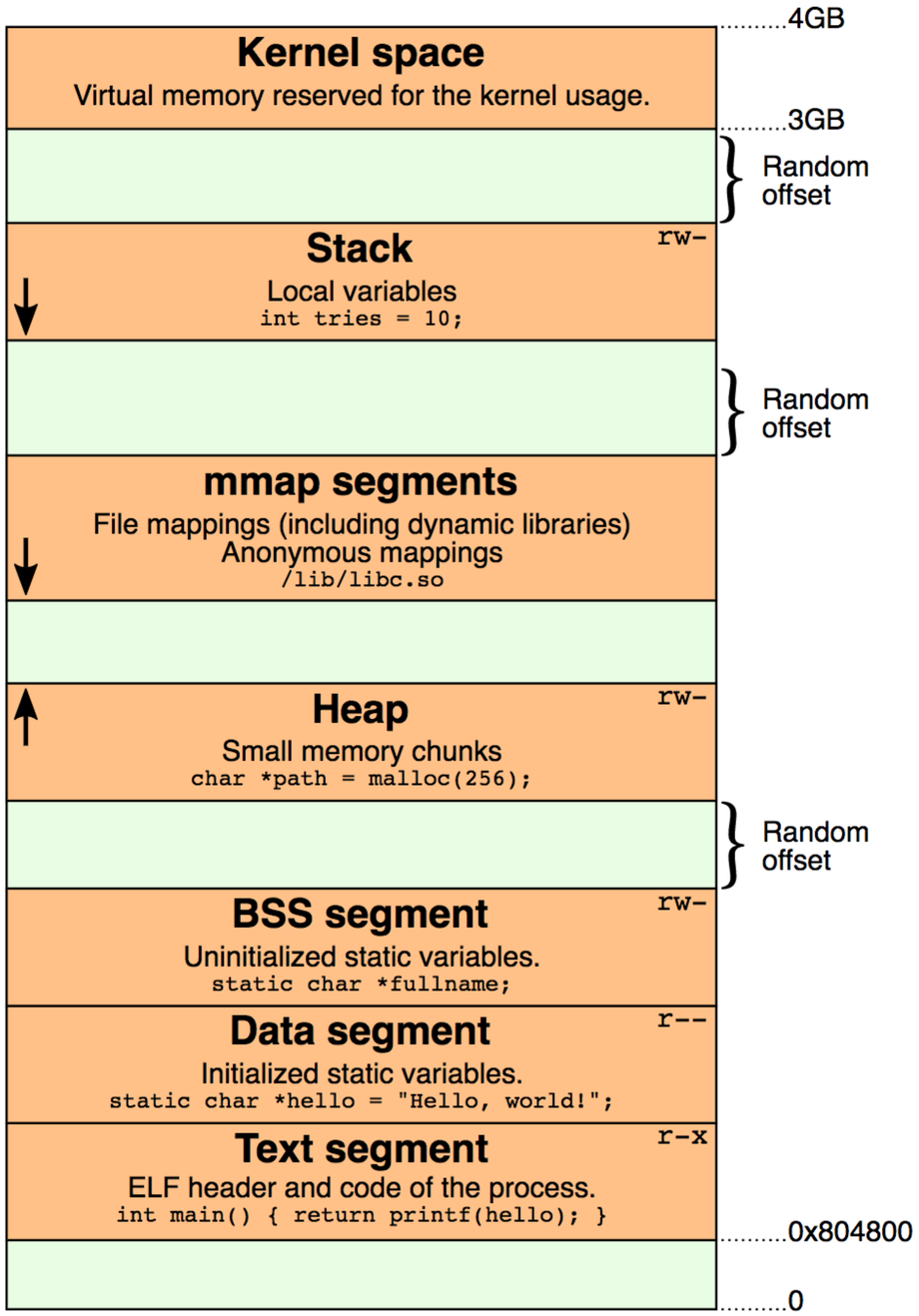
ENTRY IS CALLED
SYSCALLS[GENERIC SERVICES] ARE ACCESSED VIA:
- SYSCALL NUMBER IN THE R7 REGISTER[ARM ONLY]
- CALLING INSTRUCTION SVC

TRIVIA

THE ELF WAS FIRST SPECIFIED BY U.S.L.[UNIX SYSTEM LABORATORIES] AND U.I.[UNIFIED INTERNATIONAL] FOR UNIX SYSTEM V, IN 1989

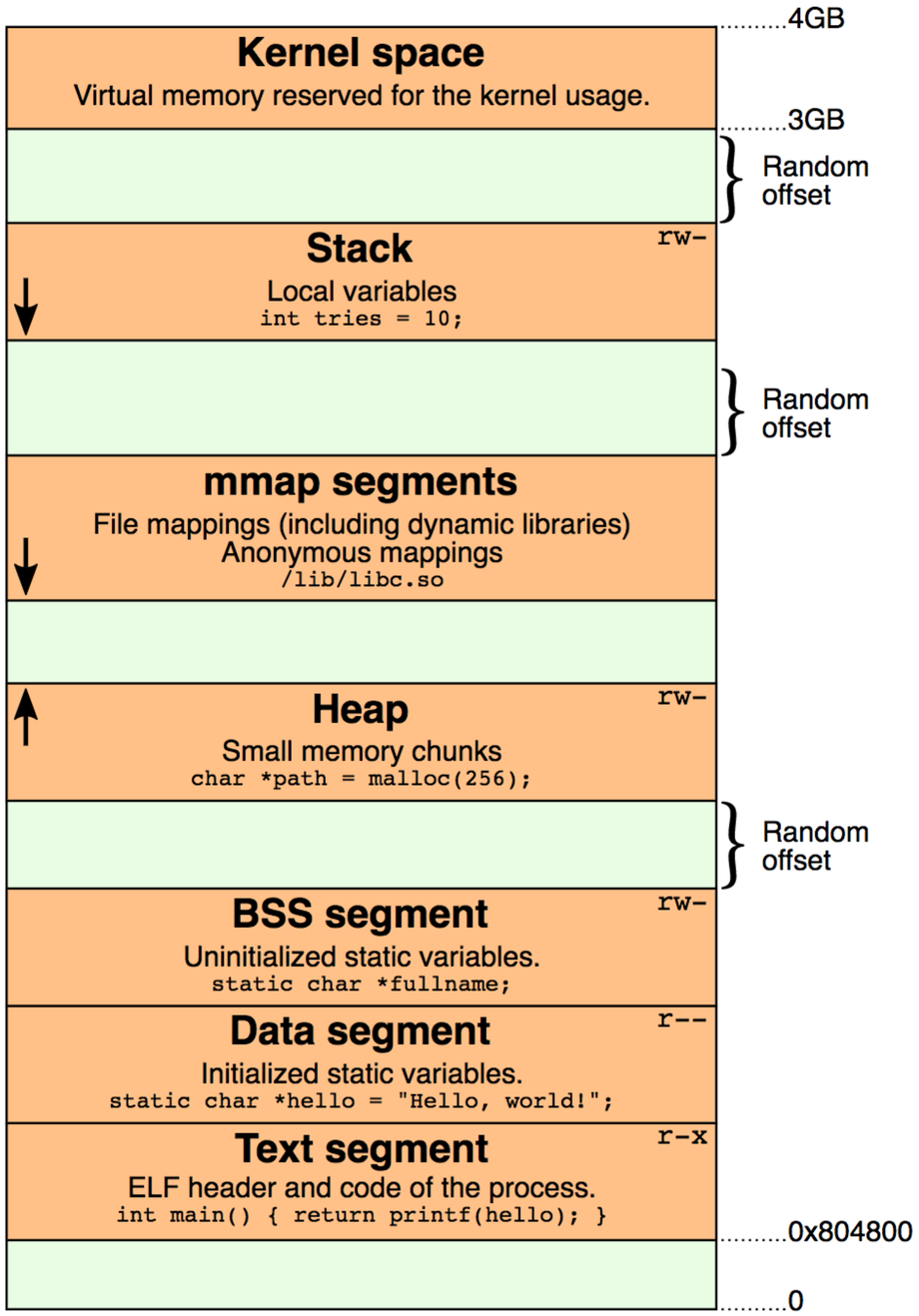
THE ELF IS USED, AMONG OTHERS, IN:
- LINUX, ANDROID, *BSD, SOLARIS, BEOS
- PSP, PLAYSTATION 2-4, DREAMCAST, GAMECUBE, WII
- VARIOUS OSES MADE BY SAMSUNG, ERICSSON, NOKIA,
- MICROCONTROLLERS FROM ATMEL, TEXAS INSTRUMENTS

VERSION 1.0A
2013/12/06



Kernel memory

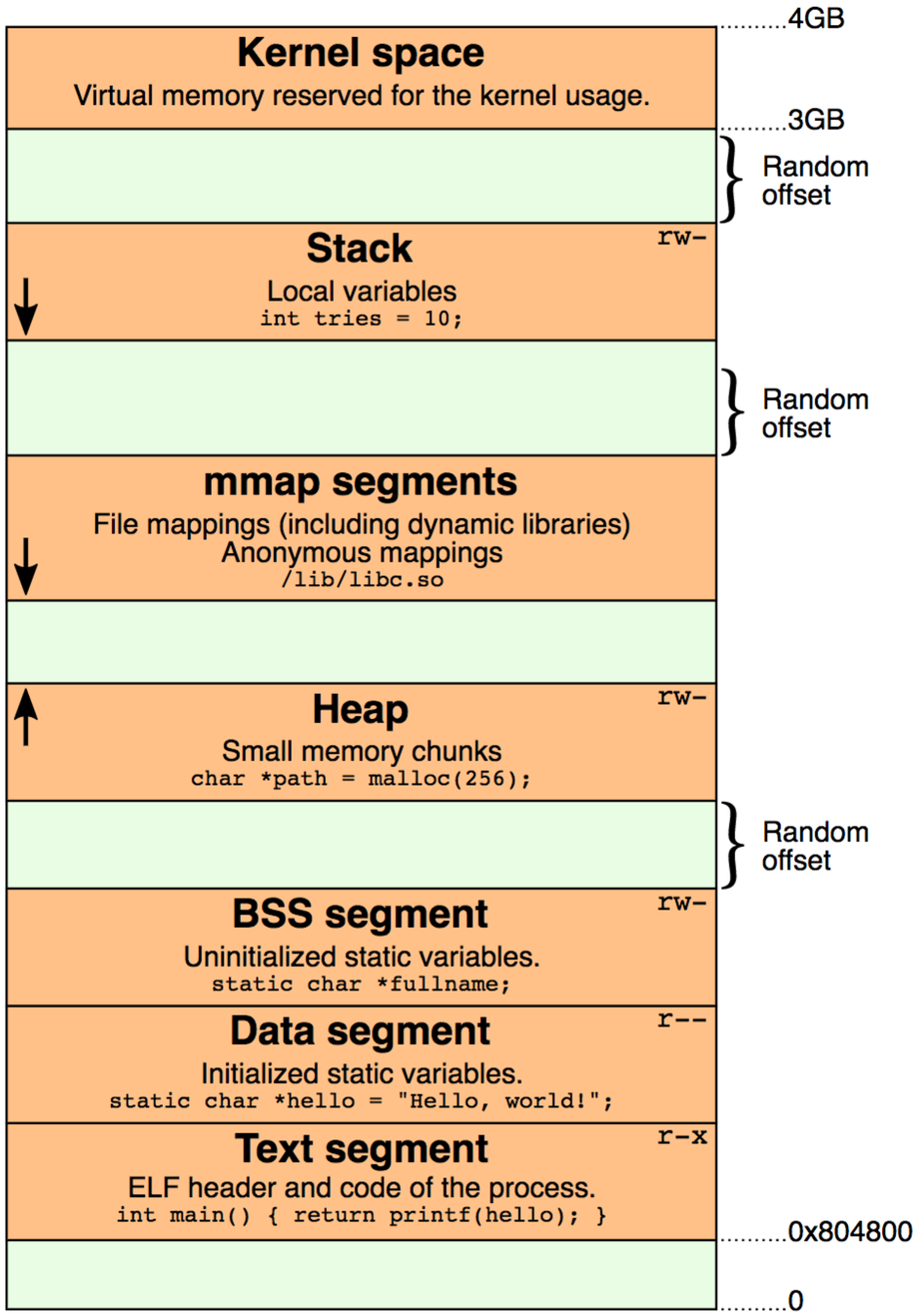
Your OS uses it



Stack: push / pop

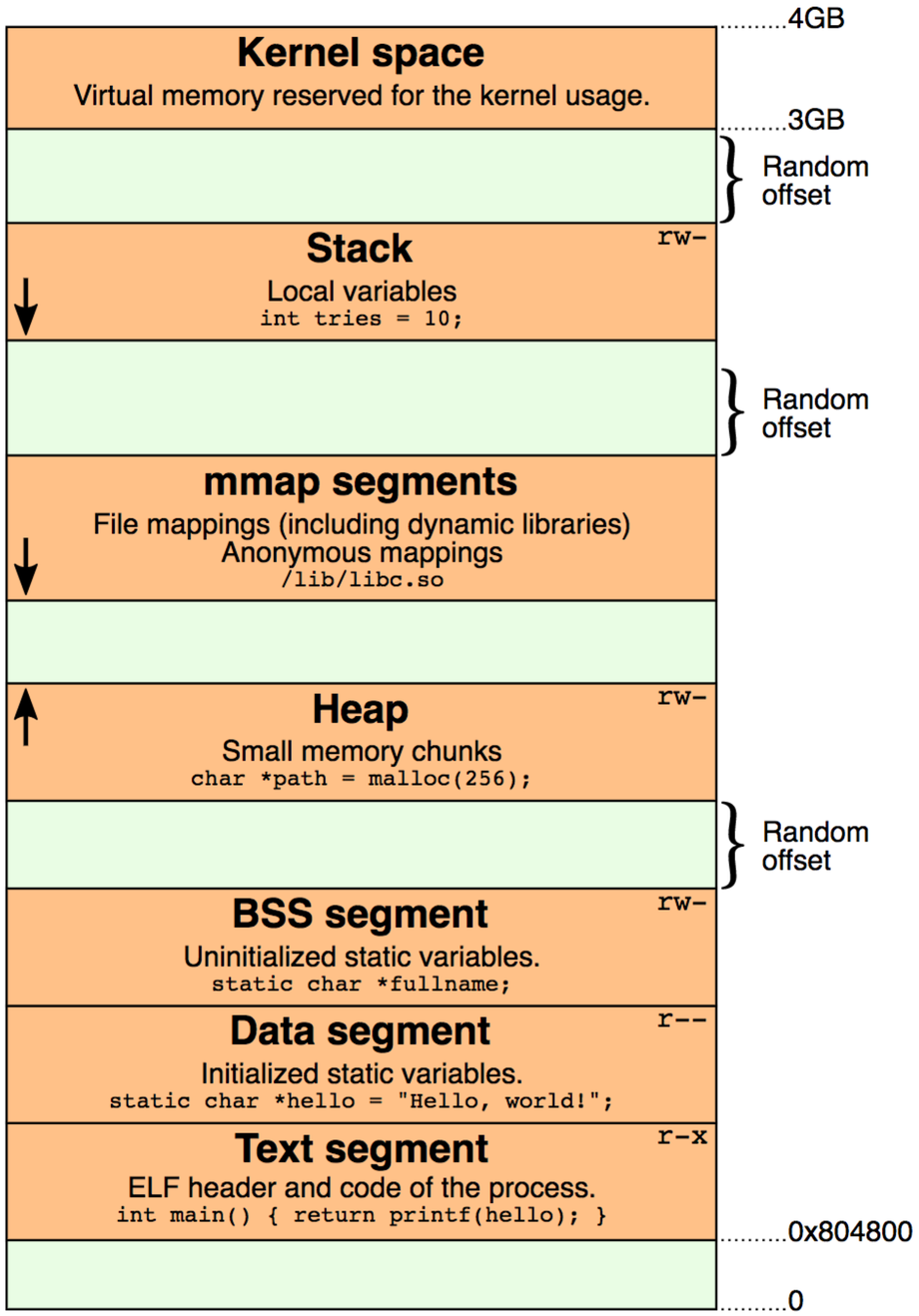
Very important:

The stack grows **down**



mmap segments

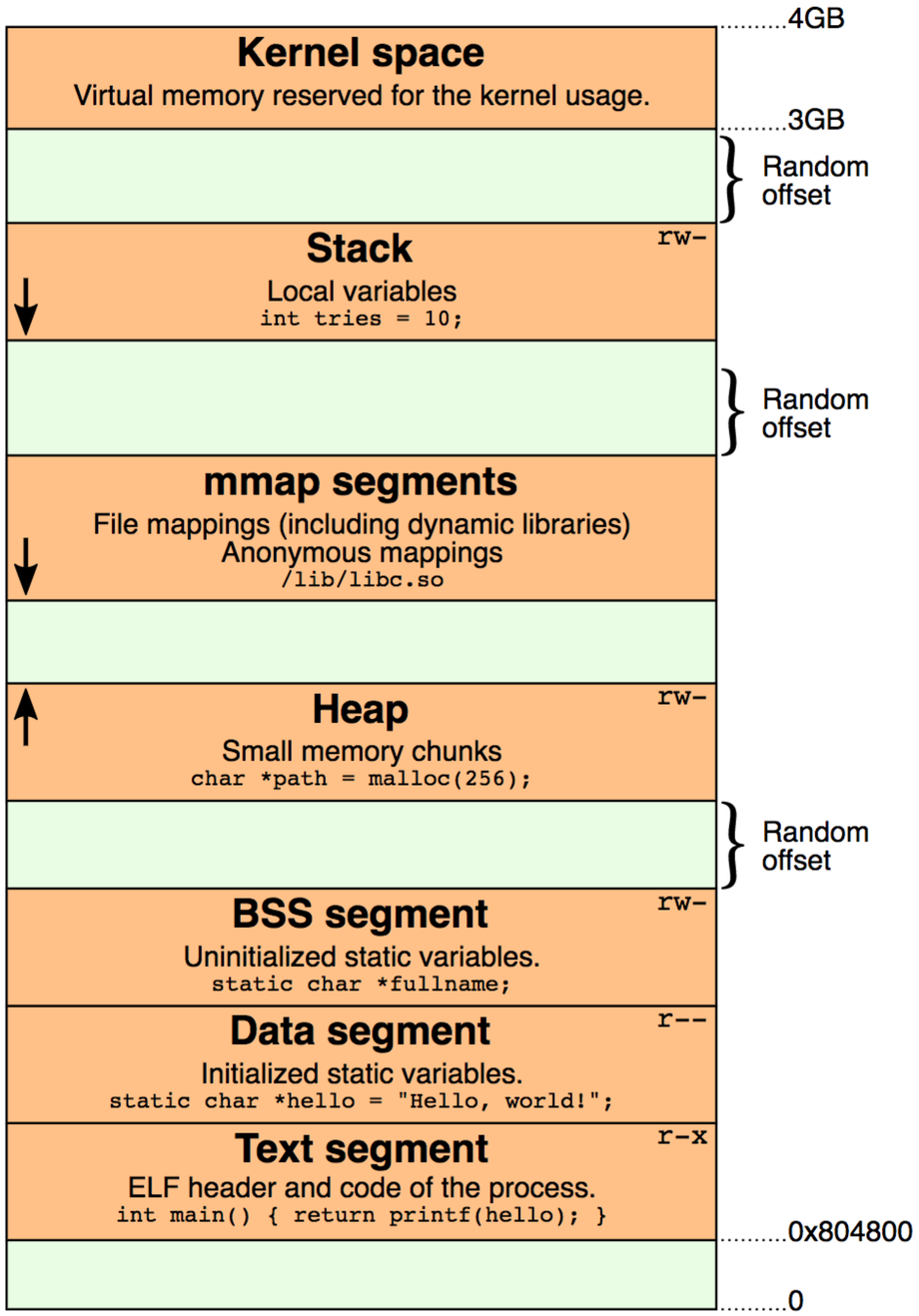
Allows you to **map** a file to memory



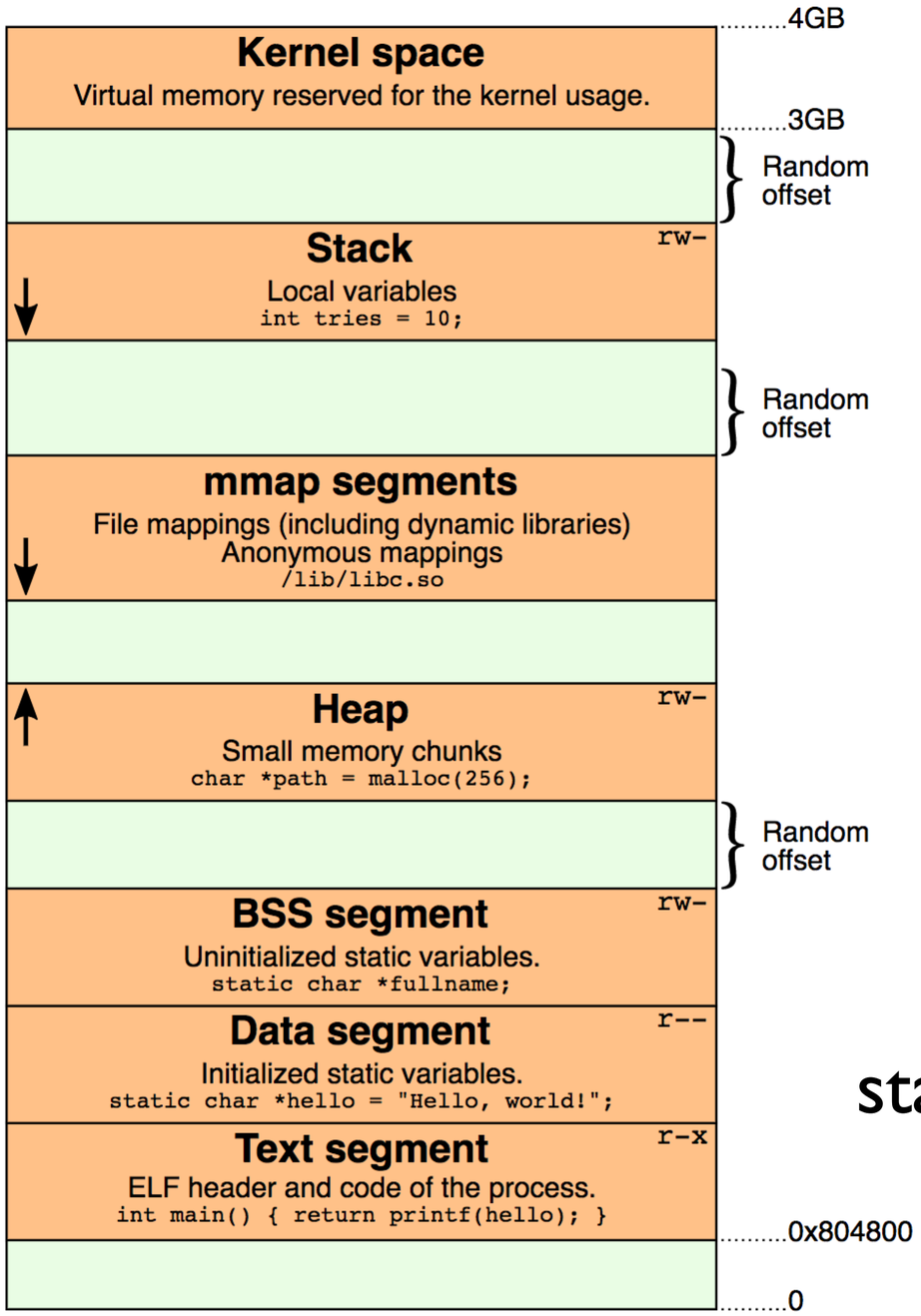
Heap: dynamic allocation

C++: New / delete

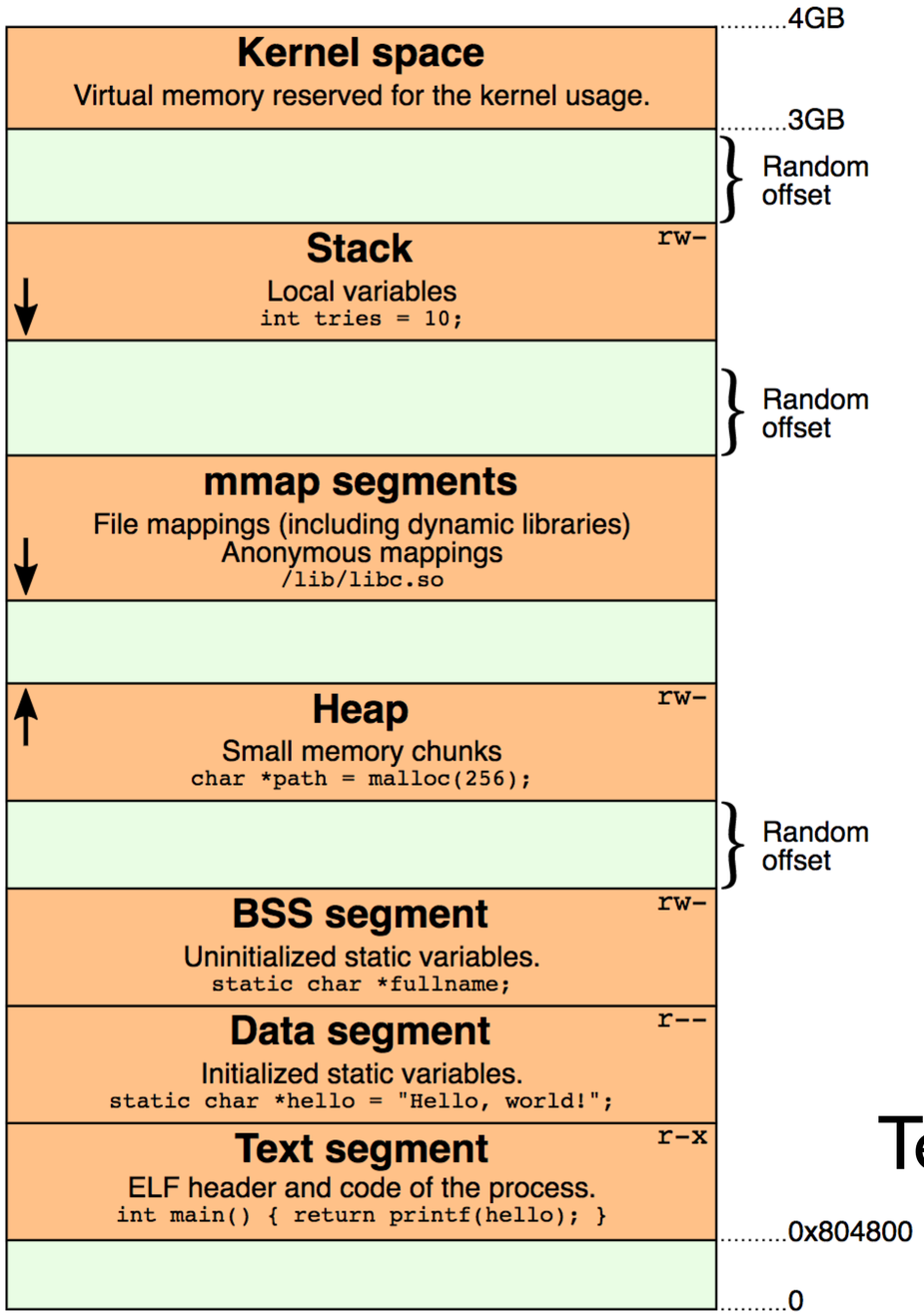
C: Malloc / free



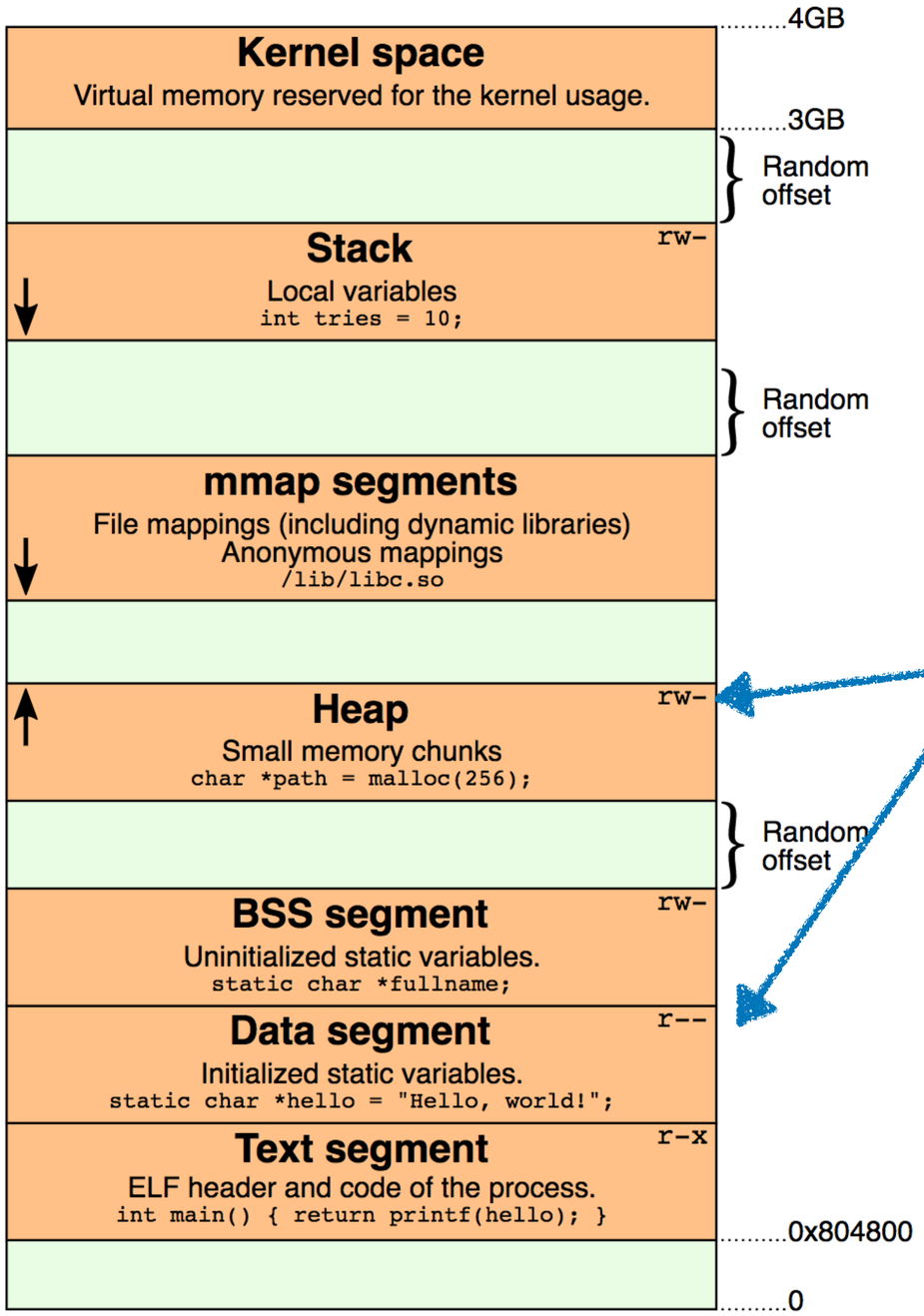
BSS: Uninitialized static vars (globals)



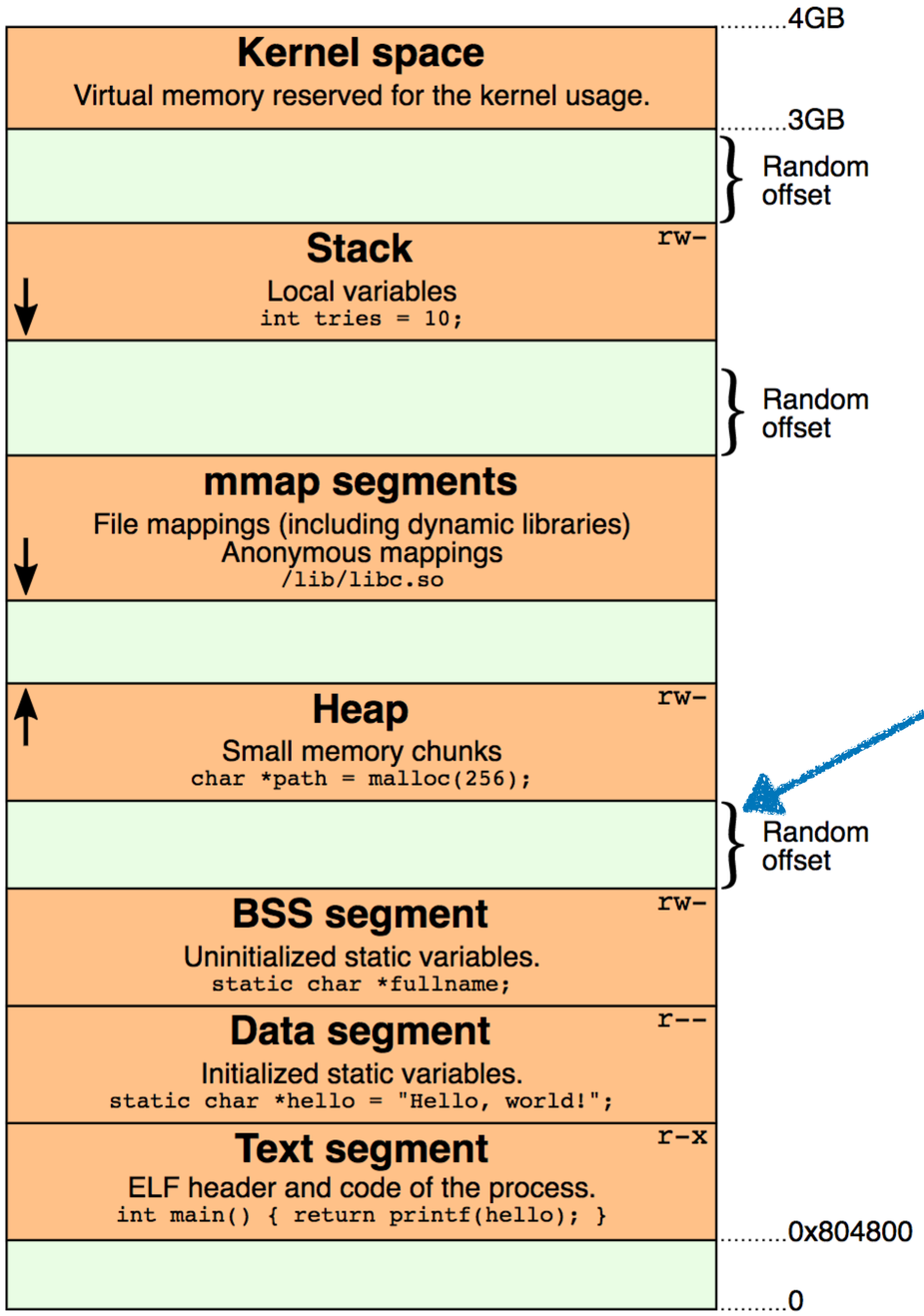
Data segment: initialized statics—e.g., constant strings



Text segment: program code



Note the **permissions**



This **random offset**
really security feature

What is the language, really?

Programs in our language are *straight-line blocks of arithmetic*:

```
x = read_int()
y = 2*x+3
z = x + y*y
return z+1
```

```
(let ([x (read)])
  (let ([y (+ (* 2 x) 3)])
    (let ([z (+ x (* y y))])
      (+ z 1))))
```

Same program, ***just different syntax***

Compiling to x86

- We look at our source code, and we look at x86
- We ask: what are the differences?
  - First step: rewrite this in x86.
- Let's carefully walk through this:
  - What are the differences?

```
(program (let ([a 1]) a))
```

```
.globl _main
.extern _read_int64
.extern _print_int64
_main:
    pushq %rbp
    movq %rsp, %rbp
    addq $-16, %rsp
    movq $1, -8(%rbp)
    movq -8(%rbp), %rax
    movq %rax, %rdi
    call _print_int64
    movq $0, %rax
    movq %rbp, %rsp
    popq %rbp
    ret
```

Key differences of concern (Sec 2.3)

- (a) x86 arithmetic instructions update their "result" argument in place, this means we can't implement x := y + z (or similar) directly
- (b) in x86, our arguments must be ***atomic*** (variables or immediate values)
  - By contrast, in LVar, they can be arbitrary nested expressions!
  - We'll need to do some flattening:
    - (+ x (+ y z)) can be written as (let ([i0 (+ y z)]) (+ x i0))
- (c) x86 has ***explicit*** control-flow: a block of instructions executing sequentially
  - LVar has an ***implicit*** control-flow, we need to make control-flow explicit
- (d) LVar can use any number of variables—only 16 registers in x86
  - Easy hack: use the stack instead
- (e) Variable names can be "shadowed," but registers / stack can't be in x86

The Big Idea

To get from source language to target language…

- Identify a set of *passes*, each of which takes *complex* features and desugars those features into more *primitive* (closer to the machine) features
- The interface between two passes constitutes an ***intermediate representation (IR)***
- *Questions*: can passes have the same input and output IR? Can passes ever be run in a loop (when might you want that)?

The Passes...

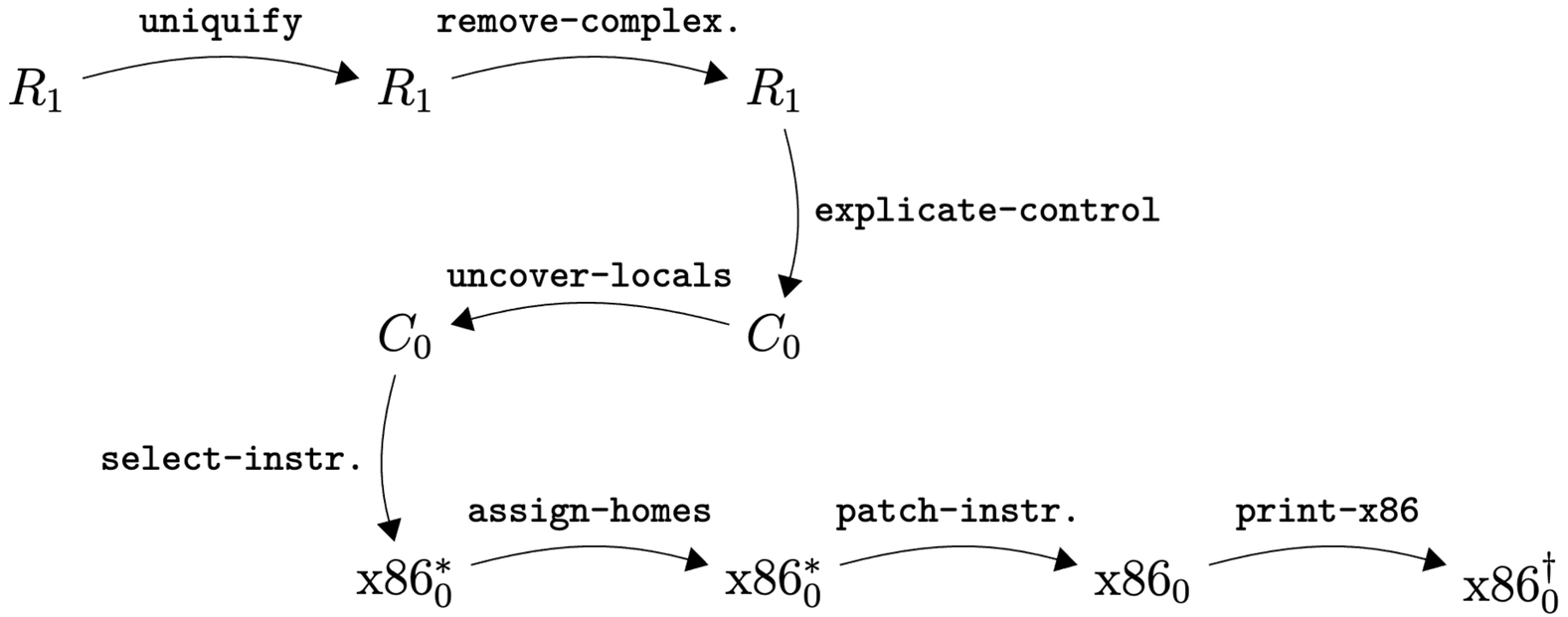


Figure 2.8: Overview of the passes for compiling R_1.

Let's brainstorm how to transform a program into x86_64 gradually

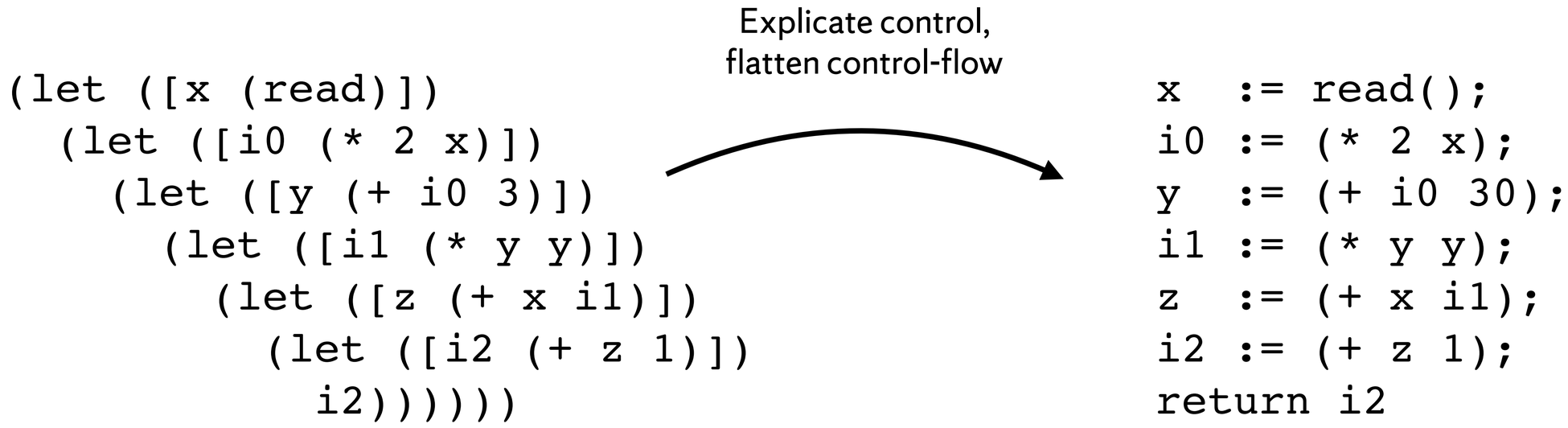
- First insight: every operation in x86_64 needs to have ***simple*** arguments
  - Can't have (+ (* x y) z), need (let ([i (* x y)]) (+ i z))
- Transform the program so that *complex* expressions are broken down into chains of simpler ones by introducing ***administrative*** bindings
- This process is called A-normal form conversion (or A-normalization)
  - The book calls it remove-complex-operand (-ator)

```
(let ([x (read)])
  (let ([y (+ (* 2 x) 3)])
    (let ([z (+ x (* y y))])
      (+ z 1))))
```

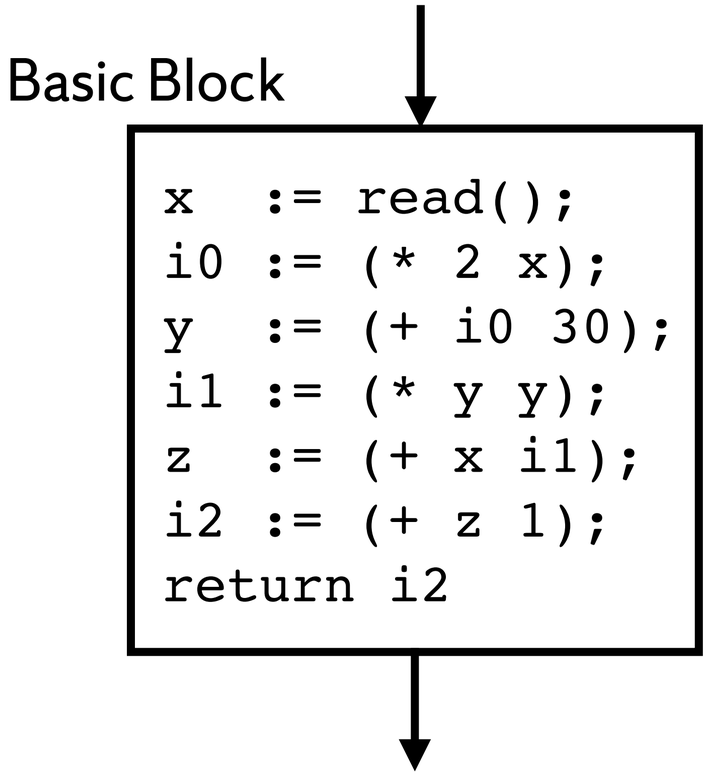
ANF

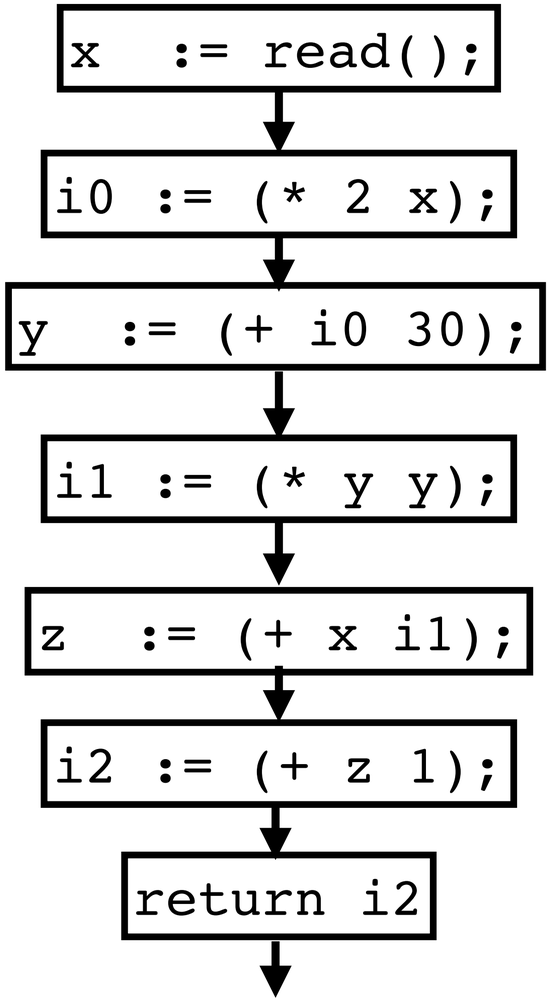
```
(let ([x (read)])
  (let ([i0 (* 2 x)])
    (let ([y (+ i0 3)])
      (let ([i1 (* y y)])
        (let ([z (+ x i1)])
          (let ([i2 (+ z 1)])
            i2))))))
```

- Now we have a really ugly nested web of expressions.
- The key is to recognize that it is not an arbitrary tree—it is a ***sequence***
- So in fact, we can just smush it together and linearize it
  - This will become more complex when we deal with branching control-flow
  - For now, things are easy—let's not make them complicated!
- The book calls this ***explicate-control***
  - Also adds the explicit return statement



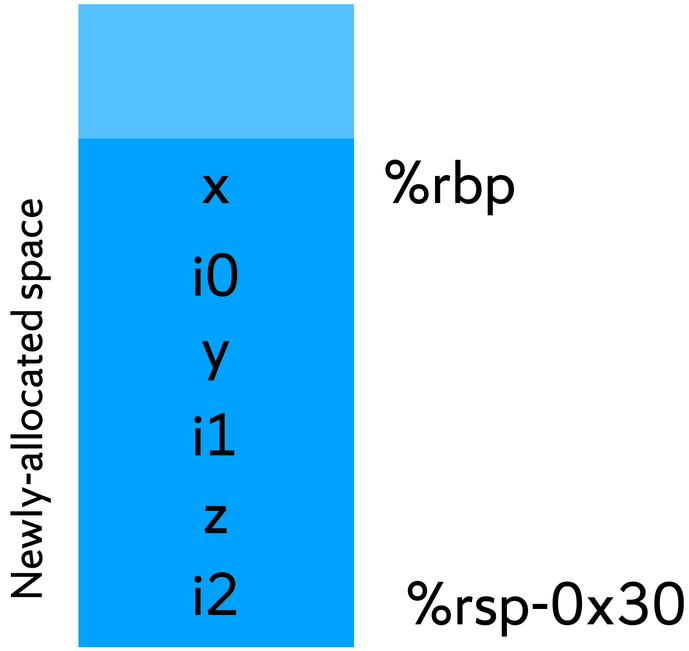
- In general, we will want to talk about a **control-flow graph**, to enable the potential for **branching** (and also *joining* after diverging)
- We can view a linear sequence as a kind of degenerate graph, where each line is a node and each node links to the next
- A straight line sequence of code is called a **basic block**





- Now that we have something like this, we're very close to assembly
- Need to allocate everything to a register and go
  - Won't work in general—we'll run out of registers
  - Let's make things dirt-simple—allocate space on the stack for each intermediate value
  - Left-hand-side of each assignment then implicitly writes to that stack location
- To do this, ***collect up all local variables*** and assemble a map of each local to a home on the stack, which we can then use to assign them…

```
x  := read();        # -8(%rbp)
i0 := (* 2 x);       # -16(%rbp)
y  := (+ i0 30);     # -24(%rbp)
i1 := (* y y);       # -32(%rbp)
z  := (+ x i1);      # -40(%rbp)
i2 := (+ z 1);       # -48(%rbp)
return i2
```



- Now we'll do ***instruction assignment***
  - Each top-level complex expression maps to (multiple) assembly instructions
  - Whenever a result gets left in a register (that should be in a variable), we move it
  - (Note: this project doesn't use *, I just include it here…)

```
x  := read();       # -8(%rbp)
i0 := (* 2 x);      # -16(%rbp)
y  := (+ i0 30);    # -24(%rbp)
i1 := (* y y);      # -32(%rbp)
z  := (+ x i1);     # -40(%rbp)
i2 := (+ z 1);      # -48(%rbp)
return i2
```

```
# x  := read();
callq read_int
movq %rax, -8(%rbp)
# i0 := (* 2 x);
movq $2, %rax
imulq -8(%rbp), %rax
movq %rax, -16(%rbp)
# y  := (+ i0 30);
movq -16(%rbp), %rax
addq  30, %rax
movq %rax, -24(%rbp)
… # pop %rbp, reset stack…
# Leave ret value in %rax
ret
```

My Advice

- The book is excellent, and makes thoughtful design choices
- If you follow the book, it gives an exact recipe for how to do the project
- But you might figure out your own way of doing it that diverges a bit from the way the book does it (e.g., is simpler)—that is natural and expected
  - In fact, a key part of the class is experimenting with different design choices; consider reimplementing parts of projects different ways, what changes?
- ***I recommend starting from the book's approach, and I will roughly follow it in class—you can take any solution as long as your produce correct input/output combinations!***
  - Project starters ***will*** deviate from the book
    - There are multiple versions of the book, I wanted my own version :-)

- The book breaks this conversion apart into a collection of tiny passes, “nano”-passes.
- Each of these “nano” passes performs a specific job, transforming an input program to an output program

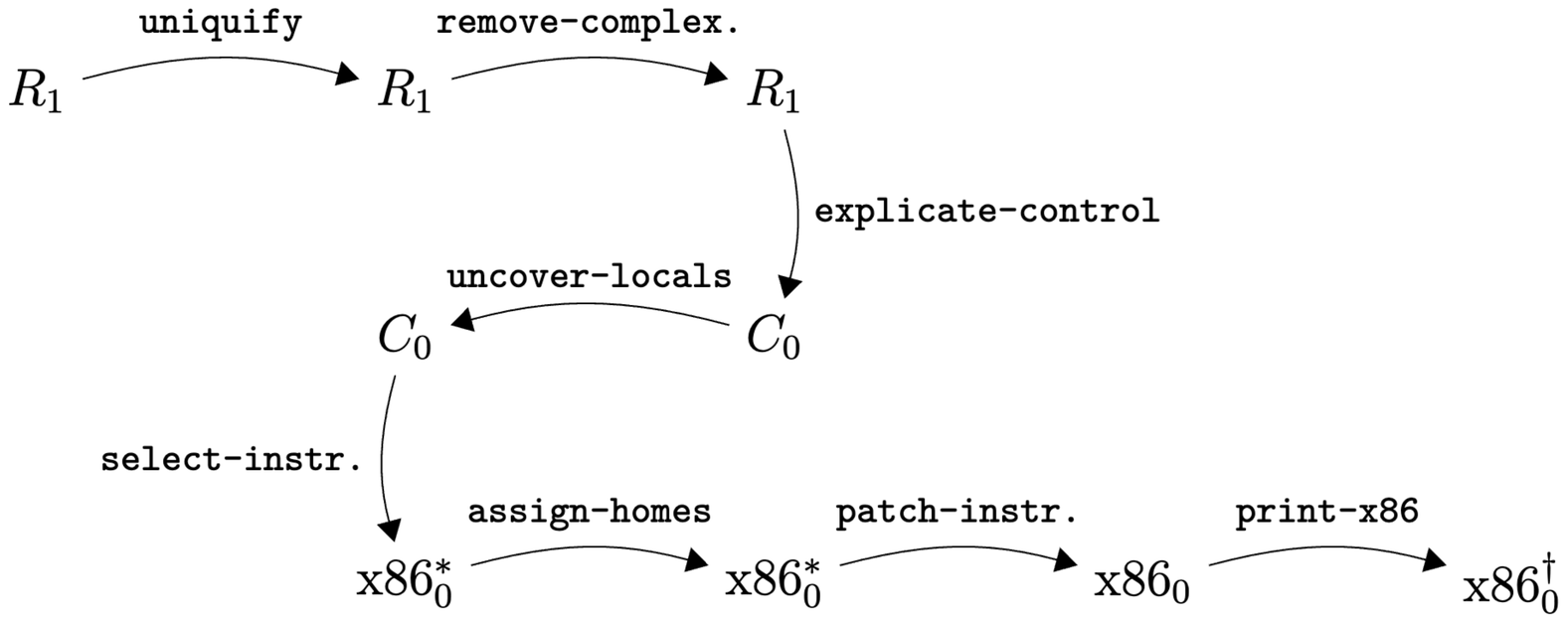


Figure 2.8: Overview of the passes for compiling R_1.

- Generally, a compiler pass will remove features
  - Take a complex feature, implement it using multiple less-complex features
  - This has pros/cons—some nuance here
- Notice how as the page goes down, the abstraction level gets lower

Closeness to machine code

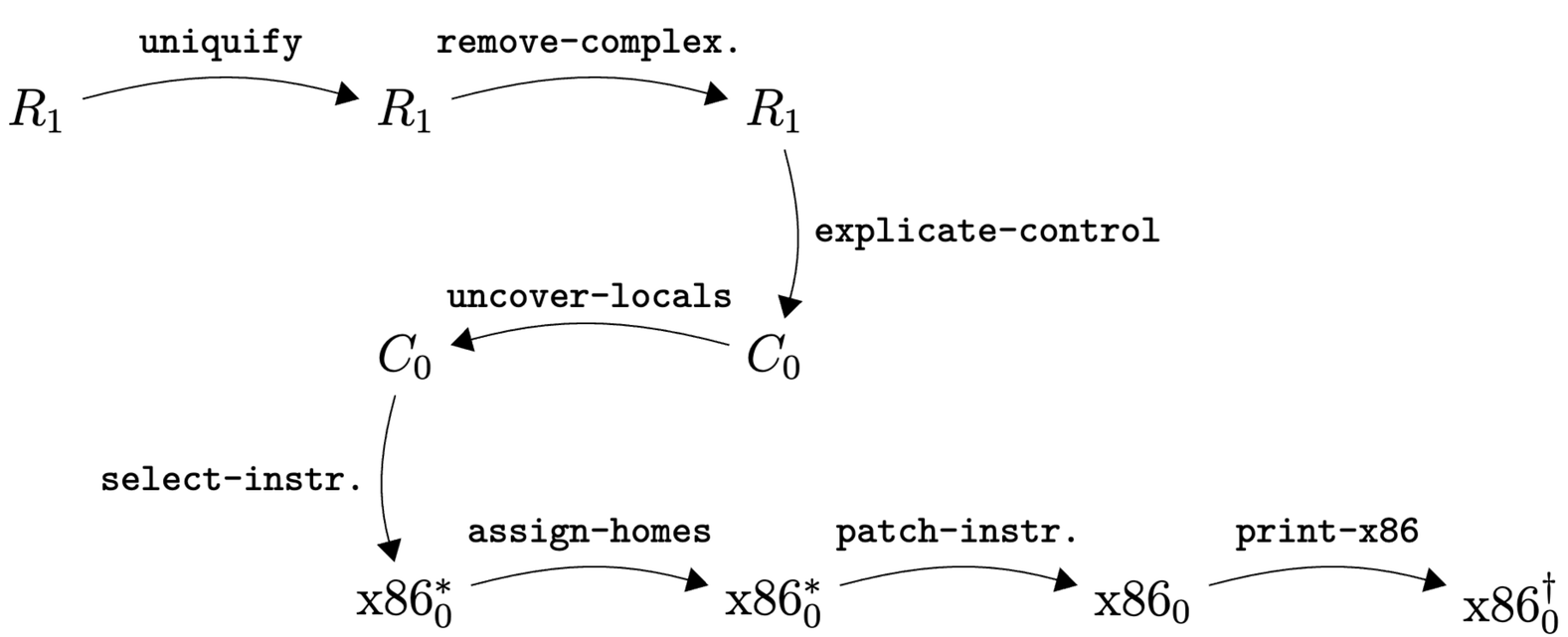


Figure 2.8: Overview of the passes for compiling R_1.

- These two passes operate on source-language code:
  - uniqueify — alpha-renames variables to avoid name-clashes
  - remove-complex — A-normalization to remove complex operations
    - Introduces administrative bindings

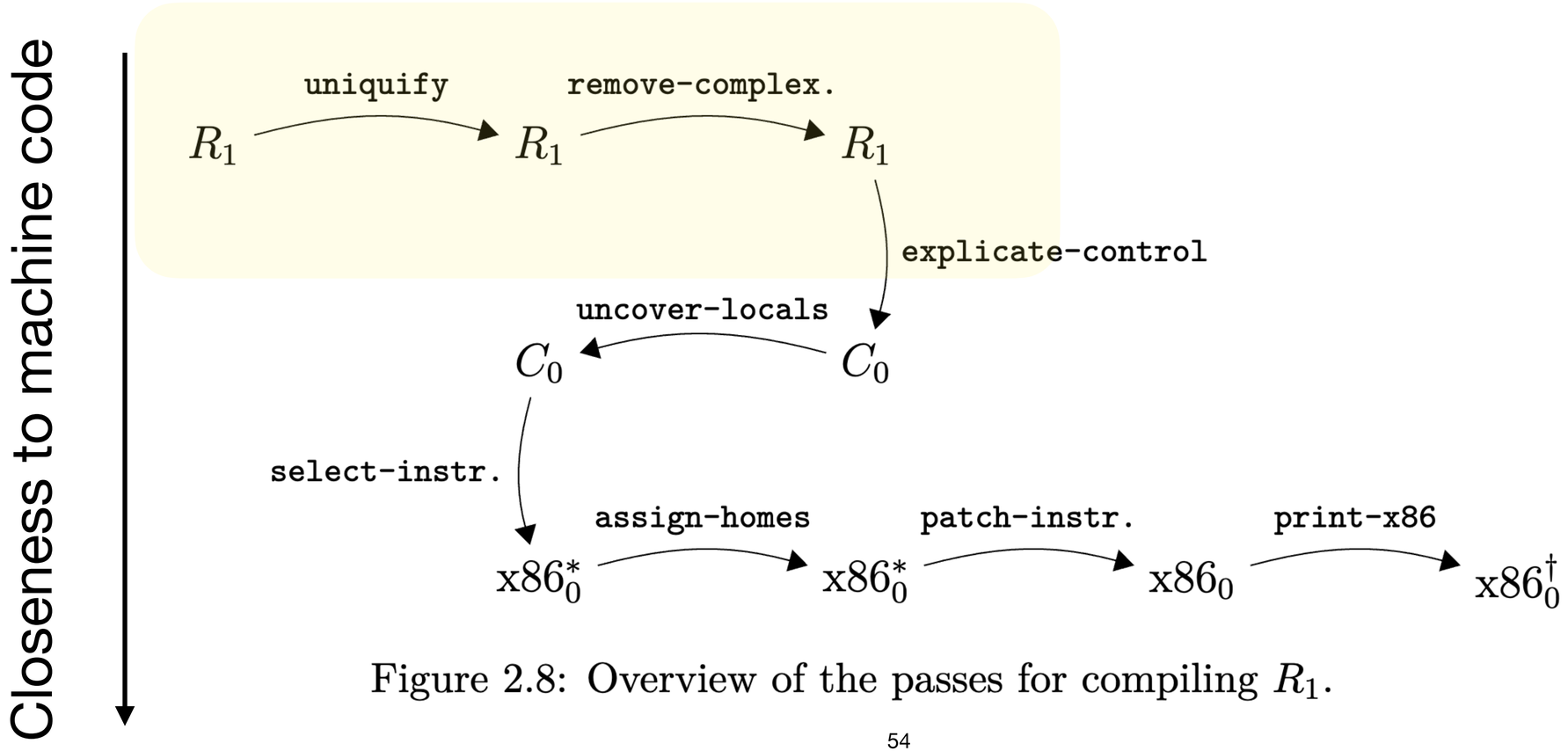


Figure 2.8: Overview of the passes for compiling R_1.

- Explicate-control finally transforms the program into an IR:
  - IR tracks a ***control-flow graph***, where program consists of **basic blocks**
  - Basic blocks—straight-line sequences of code—are ubiquitous
- uncover-locals finds out how much stack space we need

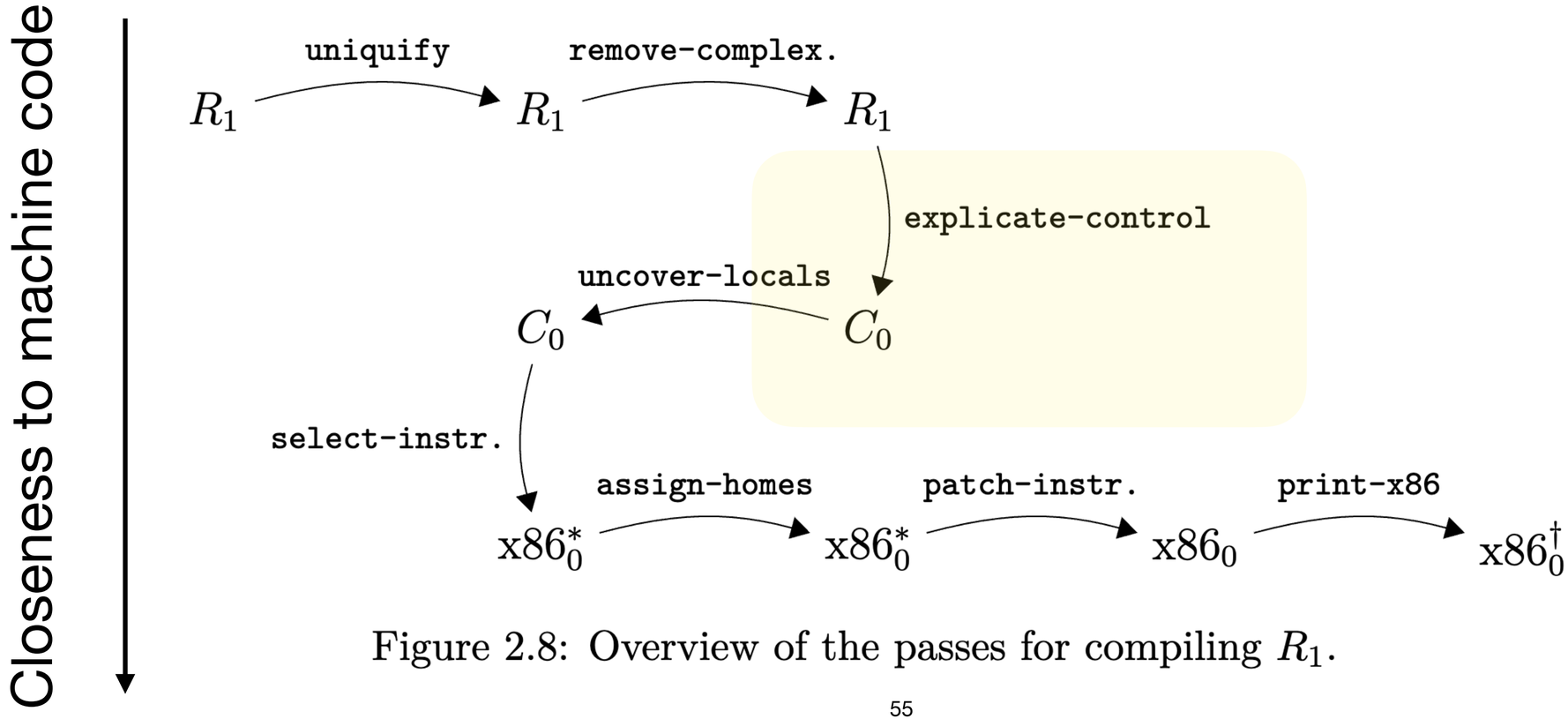


Figure 2.8: Overview of the passes for compiling R_1.

- Finally, we're working on (increasingly-lower-level variants of) x86_64:
  - Generate specific instructions, while leaving variables abstract
  - Assigning "homes" to each variable
  - Patching instructions, to ensure we adhere to addressing conventions
  - Finally, print the x86 as a block of text

Closeness to machine code

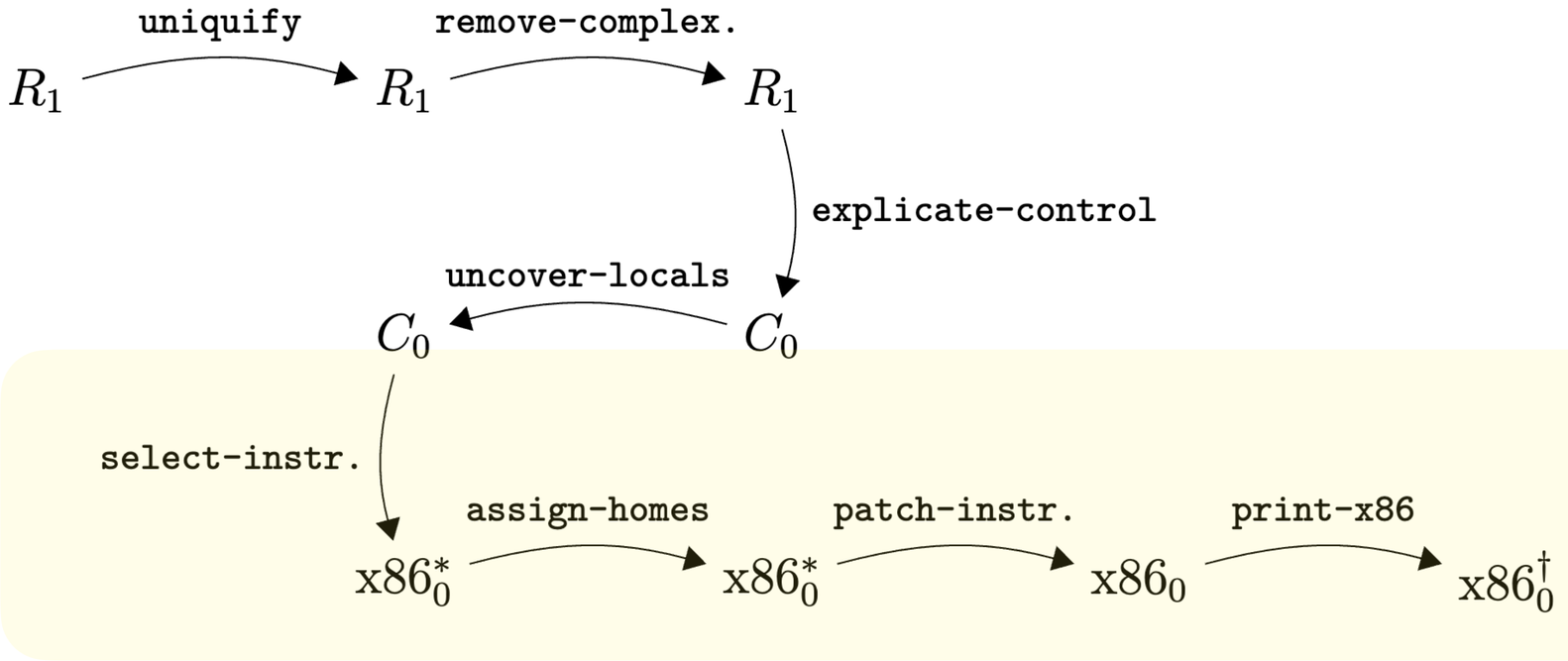


Figure 2.8: Overview of the passes for compiling R_1.

In Detail...

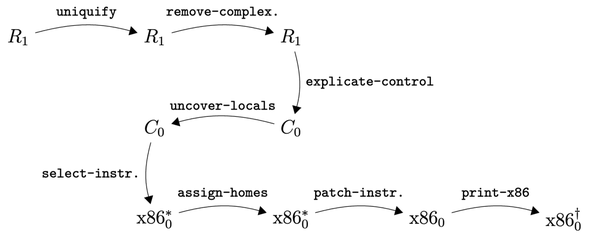


Figure 2.8: Overview of the passes for compiling R_1.

1. `uniqueify` → ensures every bound identifier is unique
   Input: `R1?` → Output: `unique-source-tree?` → Interp: `interpret-R1`
2. `anf-convert` → ANF conversion (introduce temps to flatten nested ops)
   Input: `unique-source-tree?` → Output: `anf-program?` → Interp: `interpret-anf`
3. `explicate-control` → convert ANF to C0 (a statement/block flavor)
   Input: `anf-program?` → Output: `c0-program?` → Interp: `interpret-c0`
4. `uncover-locals` → collect local variables (stack frame sizing later)
   Input: `c0-program?` → Output: `locals-program?` → Interp: `interpret-c0`
5. `select-instructions` → map C0 statements to pseudo-x86 instructions over vars
   Input: `locals-program?` → Output: `instr-program?` → Interp: `interpret-instr`
6. `assign-homes` → assign each var to a stack slot `(deref (reg rbp) offset)`
   Input: `instr-program?` → Output: `homes-assigned-program?` → Interp: `interpret-instr`
7. `patch-instructions` → break illegal memory→memory `movq` into two moves via `%rax`
   Input: `homes-assigned-program?` → Output: `patched-program?` → Interp: `interpret-instr`
8. `prelude-and-conclusion` → function prologue/epilogue; call `print_int64` on `%rax`
   Input: `patched-program?` → Output: `x86-64?` → Interp: `interpret-instr`
9. `dump-x86-64` → render to assembler text (string)
   Input: `x86-64?` → Output: `string?` → executed natively…

Uniqueify (“Alphatization”)

In Racket / Python / … we can **shadow names**. This is problematic for a compiler: let’s say you decide you want to rewrite x to be something else: now, you can’t replace x without disambiguating!

We ***alpha-convert*** unique instances of x to (gensym)’d names (“alphatization”)

```
'(program
  (let ((x (read)))
    (let ((y (let ((x (read))) (+ x (+ x x)))))
      (+ (- (+ x y)) x))))
```

```
-> unique
'(program
  ()
  (let ((x (read)))
    (let ((y (let ((x1488 (read))) (+ x1488 (+ x1488 x1488)))))
      (+ (- (+ x y)) x))))
```

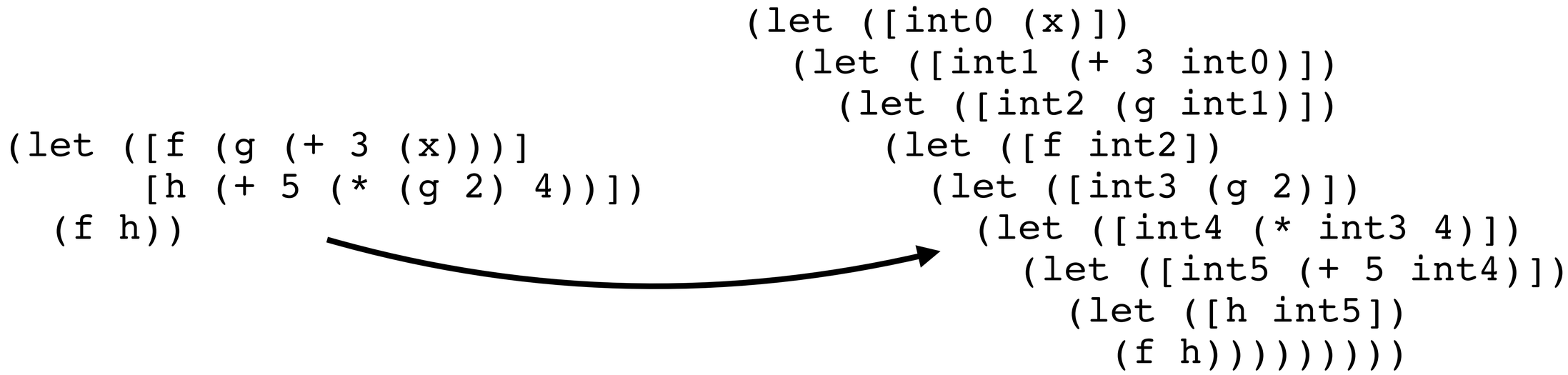
Converting to “simple” arguments

Most languages (Racket, C/C++, Java, JS) allow “complex” (i.e., nested) arguments to functions and other forms such as if, etc.

However, this is problematic for a compiler: the computer runs on a clock, and there’s no way to ensure that a complex expression can complete all in one clock cycle…

```
(let ([f (g (+ 3 (x)))]
      [h (+ 5 (* (g 2) 4))])
  (f h))
```

For example, above we have a complex expression. It is possible to rewrite it in a simpler form…



Basic idea: create an **administrative (intermediate) binding** for *each subexpression*. Use the administrative names as arguments to flatten complex expressions into sequences of “simple” expressions.

A simple expression is a function applied to atoms. An atom is either a variable or a constant.

```
(let ([f (g (+ 3 (x)))]
      [h (+ 5 (* (g 2) 4))])
  (f h))
```

```
(let ([int0 (x)])
  (let ([int1 (+ 3 int0)])
    (let ([int2 (g int1)])
      (let ([f int2])
        (let ([int3 (g 2)])
          (let ([int4 (* int3 4)])
            (let ([int5 (+ 5 int4)])
              (let ([h int5])
                (f h)))))))))
```

We will talk about this more next lecture, it is tricky and relates to a well-known compilers topic (Static Single Assignment, SSA)!

explicate-control

explicate-control is the pass that takes us from “expressions” to “blocks.” In an expression, there is nested structure based on lets, etc.

The output of explicate-control

```
-> unique
'(program
  ()
  (let ((x (read)))
    (let ((y (let ((x1488 (read))) (+ x1488 (+ x1488 x1488)))))
      (+ (- (+ x y)) x))))
```

uncover-locals

I wrote this pass for you, no need to implement it—it collects up the local variables for use in subsequent passes (e.g., when they get assigned stack locations)

patch-instructions

This pass rewrites any instructions which generate memory/memory operands

Idea: identify any operations where both arguments are in memory, and add extra **movq** instructions

prelude-and-conclusion

I have scaffolded most of this pass, along with dump-x86-64